MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

COUNTER, UP/DOWN, BINARY, DIGITAL, KEY CODE BHA

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic binary up/down counter module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.145 failures per million operating hours for class I or 0.459 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be BHA.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/292-1 for class I and M28787/292-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

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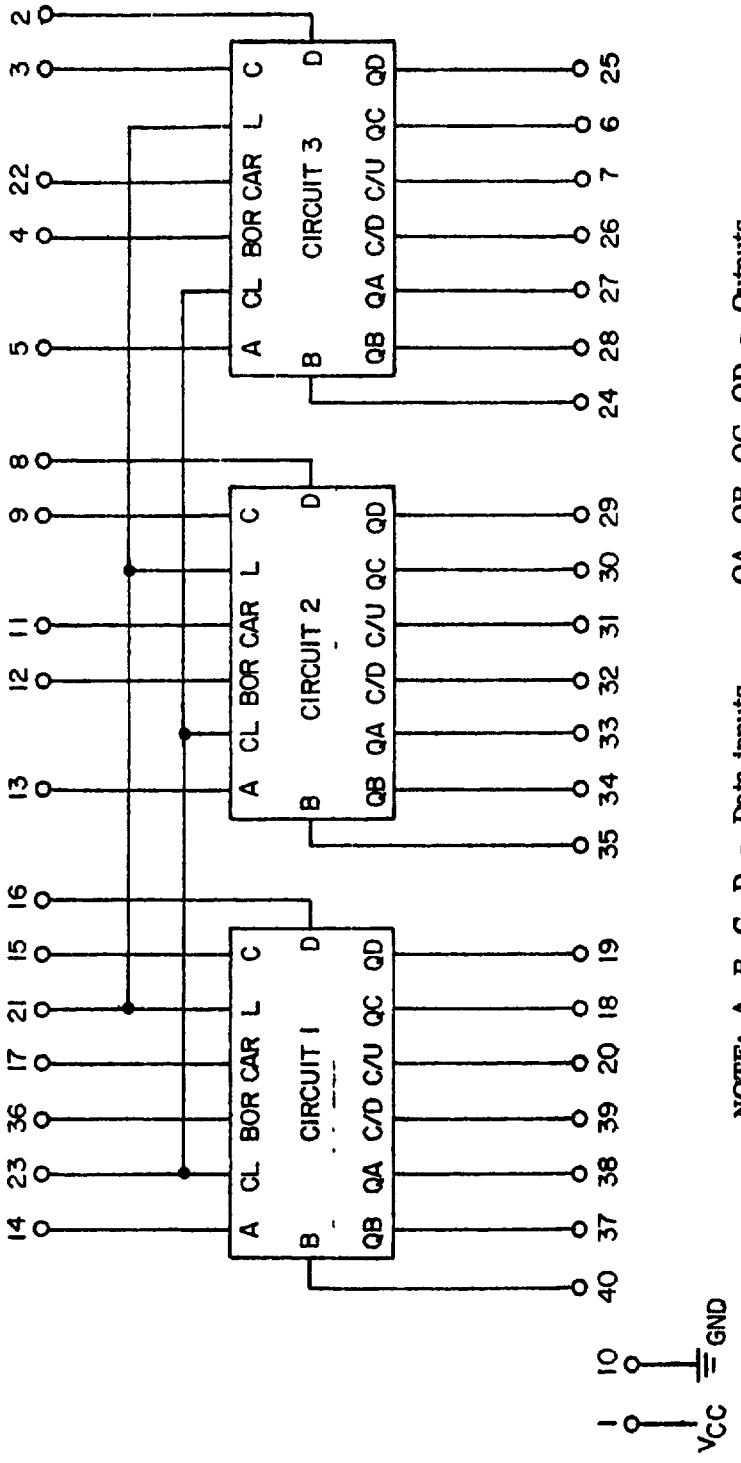


FIGURE 1. BHA module function diagram, binary up/ down counter.

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*TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation													
	Initial	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1-B18	V_{OUT} (low-loaded)	IV	S	10	800	10	800	10	800	10	800	10	800	mV
C1-C18	Input breakdown current	V	S	0	100	0	100	0	100	0	100	0	100	µA
C19, C20	Input breakdown current	V	S	0	300	0	300	0	300	0	300	0	300	µA
D1-D18	Output short circuit current	VI	X	-15	-100	-15	-100	-15	-100	-15	-100	-15	-100	mA
E1-E18	Output leakage current	VII	X	0	1	0	1	0	1	0	1	0	1	mA
F1-F20	Input clamp diode voltage	VIII	X	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
G1-G18	Output voltage, low level	IX	X	0	400	0	400	0	400	0	400	0	400	mV
H1-H18	Output voltage, high level	IX	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
J1-J53	Functional test	X	X	← See table X →										
K1-K18	Input current, low level	XI	X	-0.010	-0.435	-0.010	-0.435	-0.010	-0.435	-0.010	-0.435	-0.010	-0.435	mA
K19, K20	Input current, low level	XI	X	-0.03	-1.30	-0.03	-1 30	-0.03	-1.30	-0.03	-1.30	-0.03	-1.30	mA
L1-L18	Input current, high level	XI	X	0	20	0	20	0	20	0	20	0	20	µA
L19, L20	Input current, high level	XI	X	0	60	0	60	0	60	0	60	0	60	µA
M1	Power supply current, high level	XII	S	10	102	10	102	10	102	10	102	10	102	mA

See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
M2	Power supply current, low level	XII	S	10	102	10	102	10	102	10	102	10	102	mA
N1-N18	Transition time (t_{TLH}), C_L = 50 pF (carry, borrow and Q outputs)	XIII	S	5	50	5	50	5	55	5	50	5	60	ns
N19-N36	Transition time (t_{TLH}), C_L = 150 pF (carry, borrow and Q outputs)	XIII	Q	10	80	10	80	10	90	10	80	10	100	ns
P1-P18	Transition time (t_{THL}), C_L = 50 pF (carry, borrow and Q outputs)	XIII	S	3	18	3	18	3	20	3	18	3	22	ns
P19-P36	Transition time (t_{THL}), C_L = 150 pF (carry, borrow and Q outputs)	XIII	Q	6	40	6	40	6	42	6	40	6	44	ns
Q1-Q6	Propagation delay time (t_{PHL}), C_L = 50 pF (C/U to carry and C/D to borrow)	XIV	S	3	28	3	28	3	30	3	28	3	32	ns
Q7-Q12	Propagation delay time (t_{PHL}), C_L = 150 pF (C/U to carry and C/D to borrow)	XIV	Q	4	38	4	38	4	40	4	38	4	42	ns
R1-R3	Propagation delay time (t_{PLH}), C_L = 50 pF (C/U to carry)	XIV	S	3	30	3	30	3	32	3	30	3	34	ns
R4-R6	Propagation delay time (t_{PLH}), C_L = 50 pF (C/D to borrow)	XIV	S	3	28	3	28	3	30	3	28	3	32	ns

See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
R7-R9	Propagation delay time (t_{PLH}), C_L = 150 pF (C/D to carry)	XIV	Q	4	40	4	40	4	42	4	40	4	44	ns
R10-R12	Propagation delay time (t_{PLH}), C_L = 150 pF (C/D to borrow)	XIV	Q	4	38	4	38	4	40	4	38	4	42	ns
S1-S12	Propagation delay time (t_{PLH}), C_L = 50 pF (C/D to Q output)	XV	S	4	42	4	42	4	44	4	42	4	46	ns
S13-S24	Propagation delay time (t_{PLH}), C_L = 150 pF (C/D to Q output)	XV	Q	5	52	5	52	5	54	5	52	5	56	ns
T1-T12	Propagation delay time (t_{PHL}), C_L = 50 pF (C/D to Q output)	XV	S	5	51	5	51	5	53	5	51	5	55	ns
T13-T24	Propagation delay time (t_{PHL}), C_L = 150 pF (C/D to Q output)	XV	Q	6	61	6	61	6	63	6	61	6	65	ns
U1-U12	Propagation delay time (t_{PLH}), C_L = 50 pF (C/D to Q output)	XV	S	4	42	4	42	4	44	4	42	4	46	ns
U13-U24	Propagation delay time (t_{PLH}), C_L = 150 pF (C/D to Q output)	XV	Q	5	52	5	52	5	54	5	52	5	56	ns
V1-V12	Propagation delay time (t_{PHL}), C_L = 50 pF (C/D to Q output)	XV	S	5	51	5	51	5	53	5	51	5	55	ns
V13-V24	Propagation delay time (t_{PHL}), C_L = 150 pF (C/D to Q output)	XV	Q	6	61	6	61	6	63	6	61	6	65	ns

See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
W1-W12	Propagation delay time (t_{PLH}), C_L = 50 pF (load to Q output)	XVI	S	4	44	4	44	4	46	4	44	4	48	ns
W13-W24	Propagation delay time (t_{PLH}), C_L = 150 pF (load to Q output)	XVI	Q	5	54	5	54	5	56	5	54	5	58	ns
X1-X12	Propagation delay time (t_{PHL}), C_L = 50 pF (load to Q output)	XVI	S	4	44	4	44	4	46	4	44	4	48	ns
X13-X24	Propagation delay time (t_{PHL}), C_L = 150 pF (load to Q output)	XVI	Q	5	54	5	54	5	56	5	54	5	58	ns
Y1-Y12	Propagation delay time (t_{PHL}), C_L = 50 pF (clear to Q output and minimum clear pulse width check)	XVI	S	3	39	3	39	3	41	3	39	3	43	ns
Y13-Y24	Propagation delay time (t_{PHL}), C_L = 150 pF (clear to Q output and minimum clear pulse width check)	XVI	Q	4	49	4	49	4	51	4	49	4	53	ns
Z1-Z6	Maximum counting frequency check (minimum count pulse width check)	XVII	S	844	948	844	948	844	948	844	948	844	948	ns
AA1-AA12	Setup time (data to load) and minimum load pulse width check	XVIII	S	110	130	110	130	110	130	110	130	110	130	ns
BB1-BB12	Output crosstalk, high level	XIX	Q	0	400	0	400	0	400	0	400	0	400	mV

See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
CC1-CC12	Output crosstalk, low level	XIX	Q	0	500	0	500	0	500	0	500	0	500	mV
CC13-CC24	Output crosstalk, duration	XIX	Q	0	400	0	400	0	400	0	400	0	400	mV
DD	Power supply transient amplitude	XX	Q	0	400	0	400	0	400	0	400	0	400	mV
EE	Power supply filter capacitance	XXI	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
FF1-FF18	Input capacitance	XXII	Q	2	15	2	15	2	15	2	15	2	15	pF
FF19,FF20	Input capacitance	XXII	Q	4	30	4	30	4	30	4	30	4	30	pF
GG1-GG3	Borrow, pulse width	XXIII	Q	300	500	300	500	300	500	300	500	300	500	ns
HH1-HH24	Minimum load hold time	XXIV	S	170	230	170	230	170	230	170	230	170	230	ns

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through HH shall be performed in any order with the following exceptions. If test conditions A, B, C, D, or E are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E, where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through HH in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through EE, GG, and HH shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition EE, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through EE, GG, and HH shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through HH shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

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* 6.4 Module description. The following electrical parameters reflect the 0° to 60°C and -55° to 100°C end-of-life limits from table I.

	Class I	Class II	Unit
Power dissipation	0.561 max	0.561 max	W
Supply voltage	5.0 ±10%	5.0 ±10%	Vdc
Clock inputs			
Counting frequency	DC min, 25 max	DC min, 25 max	MHz
Pulse width	20 min	20 min	ns
Triggering edge (positive)			
Input voltage			
Low level	-0.5 min, 0.7 max	-0.5 min, 0.7 max	V
High level	2.0 min, V_{CC} max	2.0 min, V_{CC} max	V
Input current, low level			
Data, countup, countdown			
V_{IL} = 0.0 V	-0.01 min, -0.435 max	-0.01 min, -0.435 max	mA
V_{IL} = 0.4 V	-0.01 min, -0.40 max	-0.01 min, -0.40 max	mA
Clear, load			
V_{IL} = 0.0 V	-0.03 min, -1.30 max	-0.03 min, -1.30	mA
V_{IL} = 0.4 V	-0.03 min, -1.20 max	-0.03 min, -1.20	mA
Input current, high level			
Data, countup, countdown			
V_{IH} = 2.5 V	0.0 min, 20 max	0.0 min, 20 max	µA
V_{IH} = 7.0 V	0.0 min, 100 max	0.0 min, 100 max	µA
Clear, load			
V_{IH} = 2.5 V	0.0 min, 60 max	0.0 min, 60 max	µA
V_{IH} = 7.0 V	0.0 min, 300 max	0.0 min, 300 max	µA
Output current			
Low level (V_{OL} = 0.4 V)	4.0 max	4.0 max	mA
High level (V_{OH} = 2.5 V)	-400 max	-400 max	µA
Data and clear input pulse width	20 min	20 min	ns
Load input pulse width	20 min	20 min	ns
Data setup time (before positive transition of load pulse)	20 min, 22 max	20 min, 22 max	ns
Data hold time (after positive transition of load pulse)	0 min, 2 max	0 min, 2 max	ns

6.5 Module function. This module contains three 4-bit synchronous binary up/down counters. Each counter has parallel inputs and outputs and carry and borrow outputs. The counters have common clear and load inputs and independent countup and countdown clock inputs. The functional diagram is illustrated on figure 1. All three counters are fully programmable; that is, the outputs may be preset to any state by entering the desired data at the DATA inputs while the LOAD input is low. The output will change to agree with the DATA inputs independently of the count pulses. The CLEAR input forces all outputs to the low level when a high level is applied. The CLEAR function is independent of the COUNT and LOAD inputs. Both BORROW and CARRY outputs are available to cascade both the up and down counting functions. The BORROW output produces a pulse equal in width to the COUNTDOWN input when the counter underflows, and the CARRY output produces a pulse equal in width to the COUNTUP input when the counter overflows.

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6.5.1 Input to output relationship. Each counter has the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship. 1/-5/

	Inputs							Outputs			
	Clear	Load	Data A	Data B	Data C	Data D	Countup	Countdown	Q_A	Q_B	Q_C	Q_D
Asynchronous Operation	H	X	X	X	X	X	X	X	L	L	L	L
	L	L	A	B	C	D	X	X	A	B	C	D
Synchronous Operation	L	H	X	X	X	X	H	Note 1	Countdown			
	L	H	X	X	X	X	Note 1	H	Countup			

1/ Function is synchronous with the positive-going edge of the count pulse.

2/ X = Unspecified (either high level or low level).

3/ CARRY = $\overline{Q_A \bullet Q_B \bullet Q_C \bullet Q_D \bullet C/D \bullet \overline{C/U}}$ (CARRY is ACTIVE low).

4/ BORROW = $\overline{\overline{Q_A} \bullet \overline{Q_B} \bullet \overline{Q_C} \bullet \overline{Q_D} \bullet C/U \bullet \overline{C/D}}$ (BORROW is ACTIVE low).

5/ L = low level and H = high level.

6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

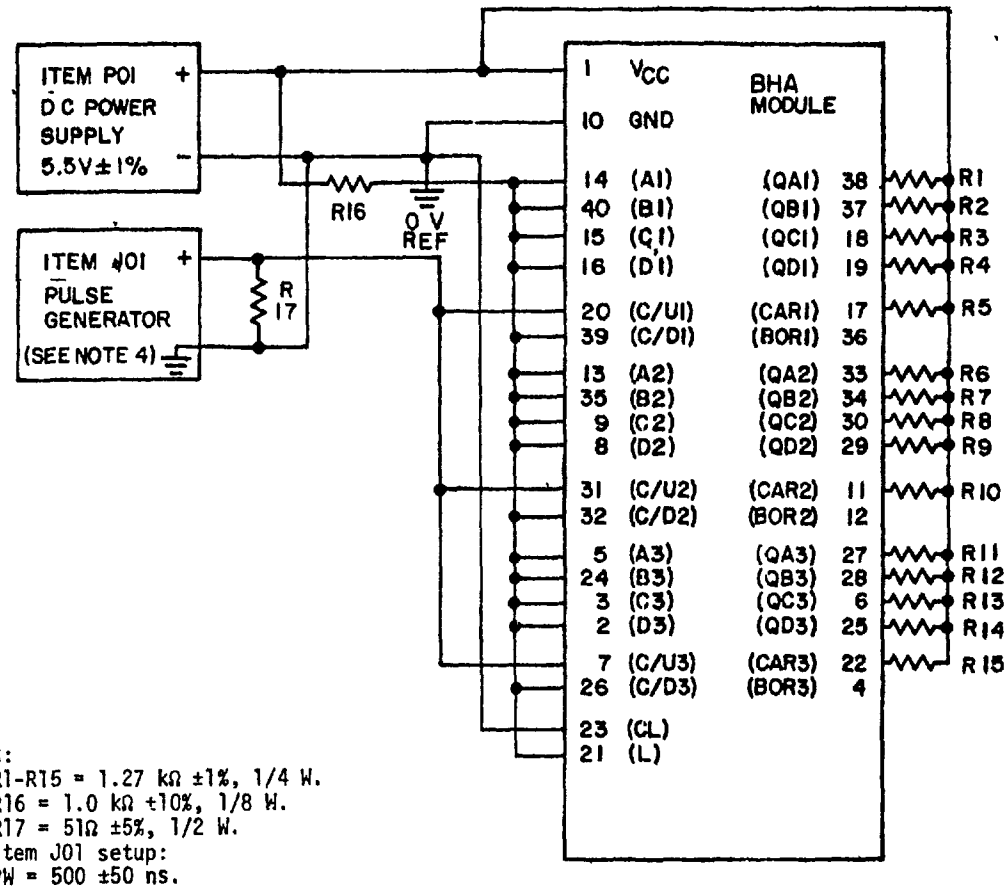
6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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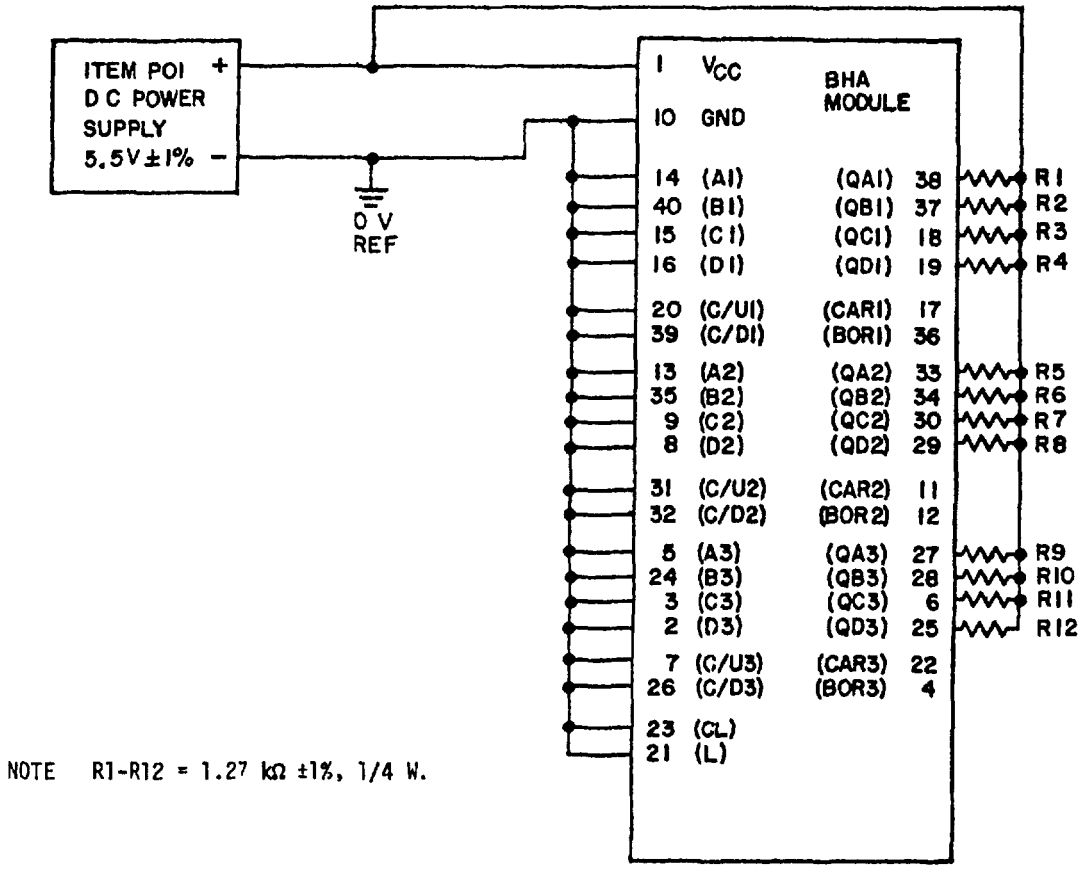
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NOTES:
1. R1-R15 = 1.27 kΩ ±1%, 1/4 W.
2. R16 = 1.0 kΩ ±10%, 1/8 W.
3. R17 = 51Ω ±5%, 1/2 W.
4. Item J01 setup:
   PW = 500 ±50 ns.
   PRP = 1.0 ±0.1 µs.
   $t_{TLH} = t_{THL}$ = 50 ±20 ns.
   PA = 4.0 ±0.2 V.

FIGURE 2. Operating temperature and life test circuit.

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NOTE R1-R12 = 1.27 kΩ ±1%, 1/4 W.

FIGURE 3. Component temperature test circuit.

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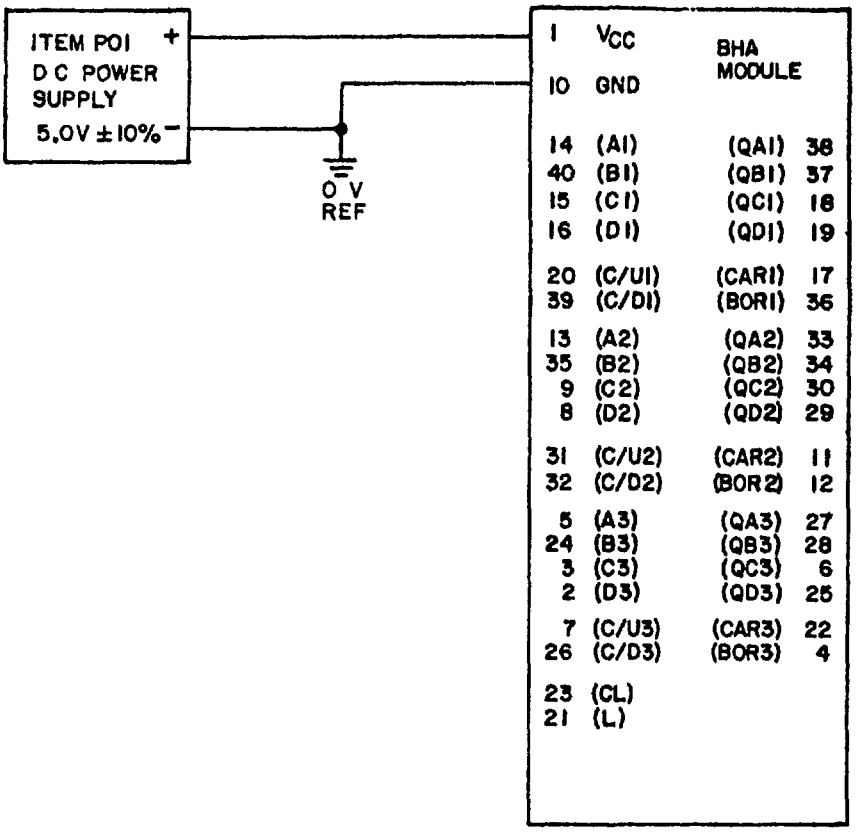


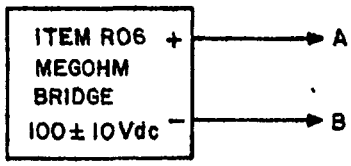
FIGURE 4. Vibration test circuit.

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TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	All pins	Frame	10	∞	MΩ	5

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1	V_{CC}	BHA MODULE	
10	GND		
14	(AI)	(QAI)	38
40	(BI)	(QBI)	37
15	(CI)	(QCI)	18
16	(DI)	(QDI)	19
20	(C/UI)	(CARI)	17
39	(C/DI)	(BORI)	36
13	(A2)	(QA2)	33
35	(B2)	(QB2)	34
9	(C2)	(QC2)	30
8	(D2)	(QD2)	29
31	(C/U2)	(CAR2)	11
32	(C/D2)	(BOR2)	12
5	(A3)	(QA3)	27
24	(B3)	(QB3)	28
3	(C3)	(QC3)	6
2	(D3)	(QD3)	25
7	(C/U3)	(CAR3)	22
26	(C/D3)	(BOR3)	4
23	(CL)		
21	(L)		

WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe.

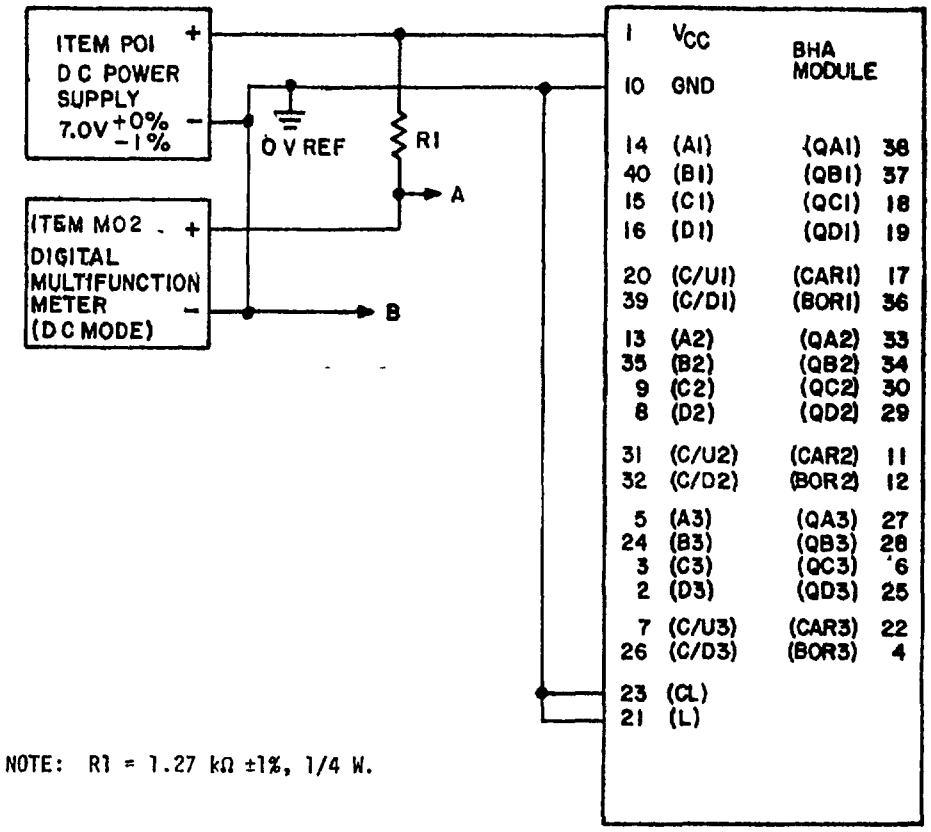
FIGURE 5. Isolation test circuit for table III.

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*TABLE IV. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect A to pin	Connect B to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
B1	QA1 V_{OUT} (low loaded)	38	14	10	800	mV	6
B2	QB1	37	40				
B3	QC1	18	15				
B4	QD1	19	16				
B5	CAR1	17	20				
B6	BOR1	36	14,15,16,39,40				
B7	QA2	33	13				
B8	QB2	34	35				
B9	QC2	30	9				
B10	QD2	29	8				
B11	CAR2	11	31				
B12	BOR2	12	8,9,13,32,35				
B13	QA3	27	5				
B14	QB3	28	24				
B15	QC3	6	3				
B16	QD3	25	2				
B17	CAR3	22	7				
B18	BOR3	4	2,3,5,24,26				

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NOTE: R1 = 1.27 kΩ ±1%, 1/4 W.

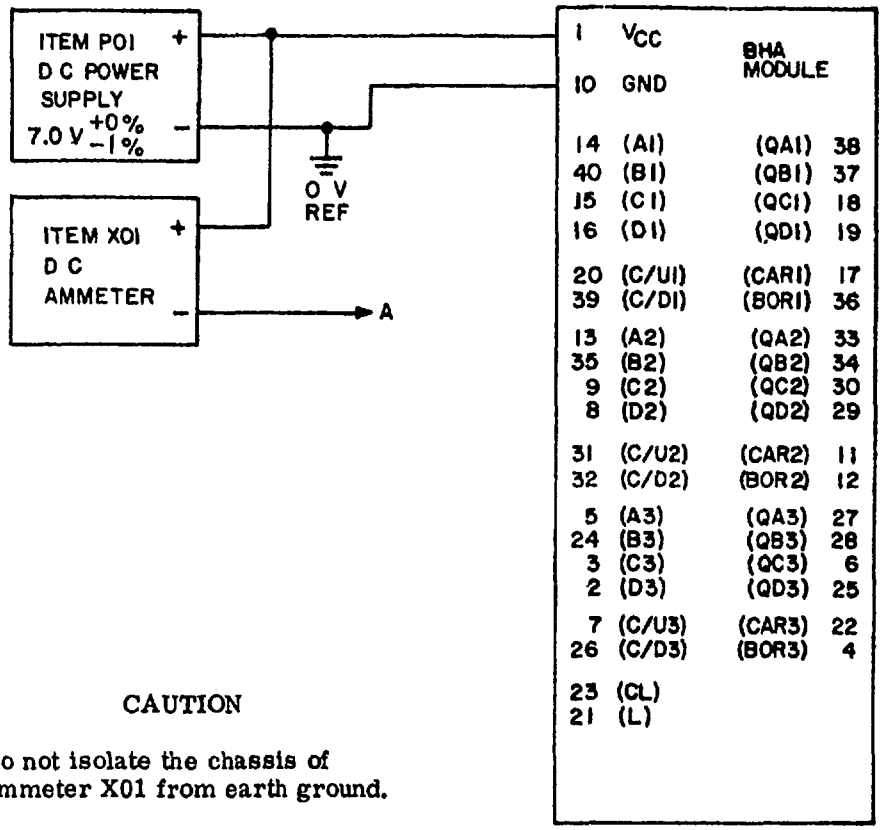
FIGURE 6. V_{OUT} (low-loaded) test circuit for table IV.

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TABLE V. Input breakdown current test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
C1	A1 Input breakdown current	14	0	100	µA	7
C2	B1	40				
C3	C1	15				
C4	D1	16				
C5	C/U1	20				
C6	C/D1	39				
C7	A2	13				
C8	B2	35				
C9	C2	9				
C10	D2	8				
C11	C/U2	31				
C12	C/D2	32				
C13	A3	5				
C14	B3	24				
C15	C3	3				
C16	D3	2				
C17	C/U3	7				
C18	C/D3	26				
C19	CL	23		300		
C20	L	21		300		

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CAUTION

Do not isolate the chassis of ammeter X01 from earth ground.

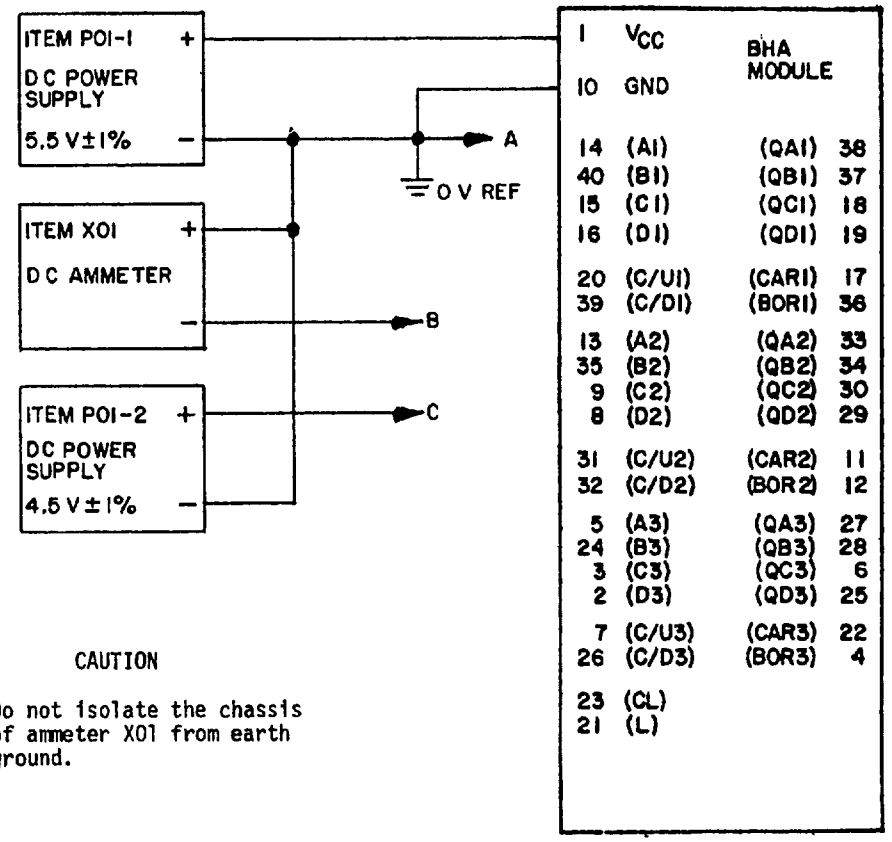
FIGURE 7. Input breakdown current test circuit for table V.

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TABLE VI. Output short circuit current.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
D1	QA1 Output short circuit current	21, 23	38	14,20,39	-15	-100	mA	8
D2	QB1		37	20,39,40				
D3	QC1		18	15,20,39				
D4	QD1		19	16,20,39				
D5	CAR1		17	20,39				
D6	BOR1		36	20,39				
D7	QA2		33	13,31,32				
D8	QB2		34	31,32,35				
D9	QC2		30	9,31,32				
D10	QD2		29	8,31,32				
D11	CAR2		11	31,32				
D12	BOR2		12	31,32				
D13	QA3		27	5,7,26				
D14	QB3		28	7,24,26				
D15	QC3		6	3,7,26				
D16	QD3		25	2,7,26				
D17	CAR3		22	7,26				
D18	BOR3		4	7,26				

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NOTE: Only one output shall be shorted at a time and the measurement taken within 10 seconds.

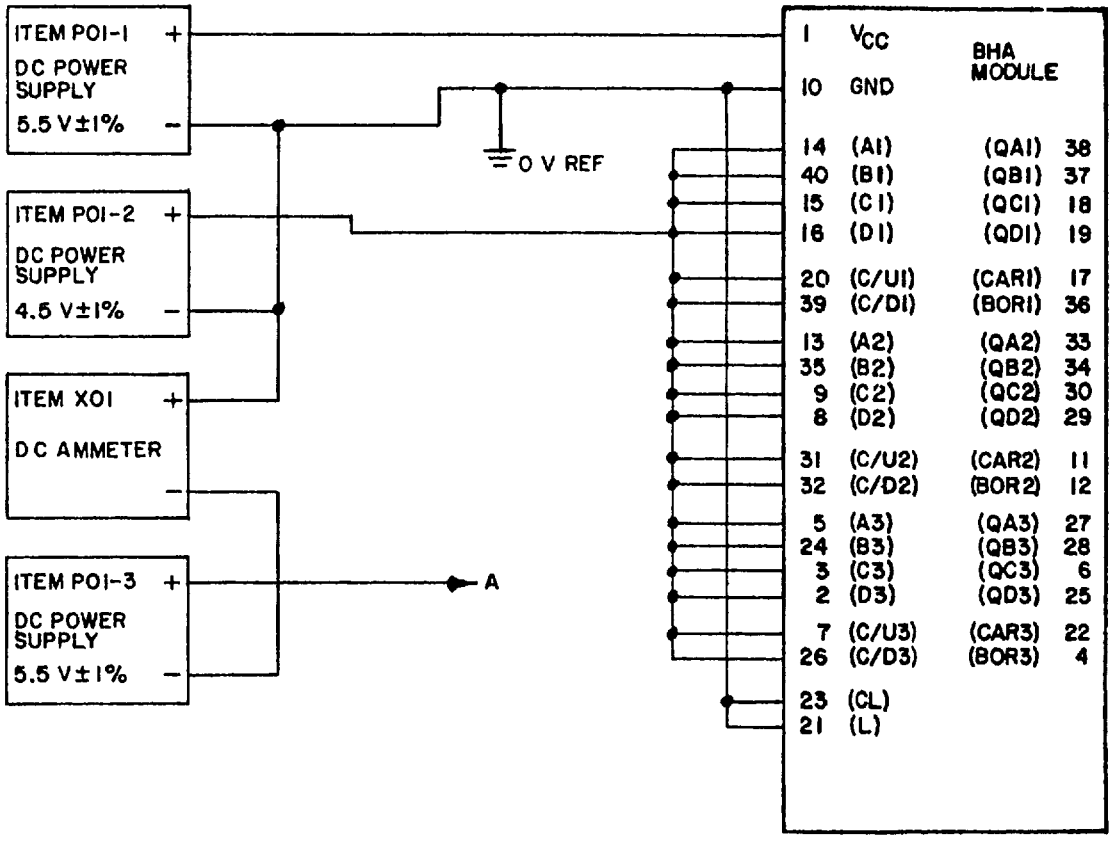
FIGURE 8 Output short circuit current test circuit for table VI.

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TABLE VII. Output leakage current test.

Test condition	Parameter		Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
E1	QA1	Output leakage current	38	0	1	mA	9
E2	QB1		37				
E3	QC1		18				
E4	QD1		19				
E5	CAR1		17				
E6	BOR1		36				
E7	QA2		33				
E8	QB2		34				
E9	QC2		30				
E10	QD2		29				
E11	CAR2		11				
E12	BOR2		12				
E13	QA3		27				
E14	QB3		28				
E15	QC3		6				
E16	QD3		25				
E17	CAR3		22				
E18	BOR3		4				

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CAUTION

Do not isolate the chassis of ammeter XOI from earth ground.

NOTE: ITEM POI-3 shall be current limited to 5.0 mA.

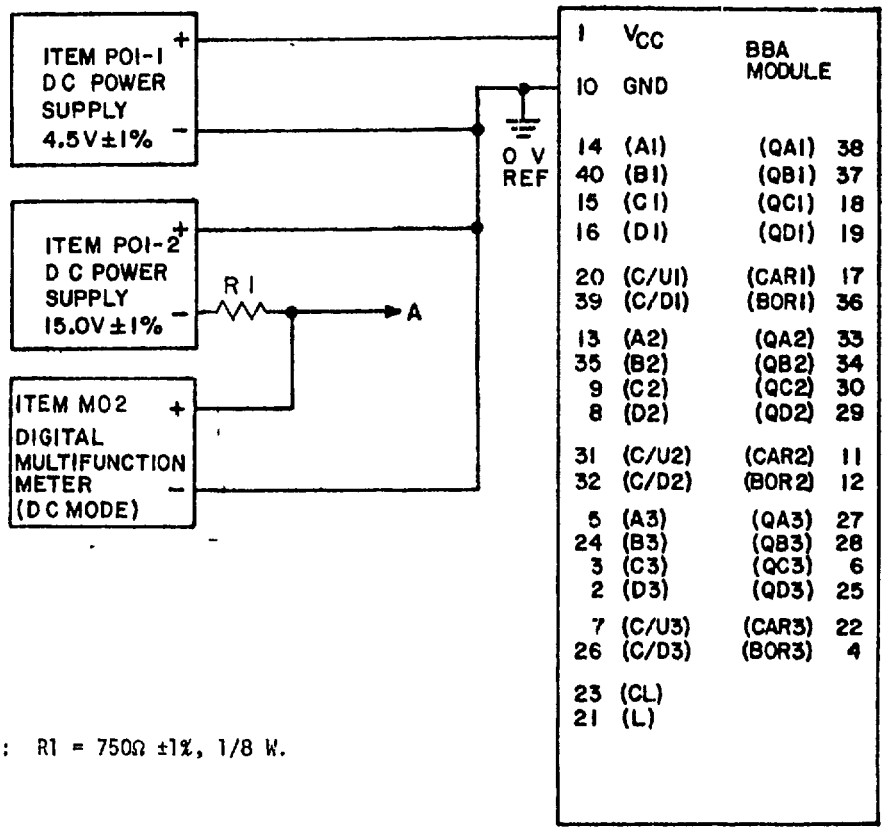
FIGURE 9. Output leakage current test circuit for table VII.

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TABLE VIII. Input clamp diode voltage test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
F1	A1 Input clamp diode voltage	14	-0.4	-1.5	V	10
F2	B1	40				
F3	C1	15				
F4	D1	16				
F5	C/U1	20				
F6	C/D1	39				
F7	A2	13				
F8	B2	35				
F9	C2	9				
F10	D2	8				
F11	C/U2	31				
F12	C/D2	32				
F13	A3	5				
F14	B3	24				
F15	C3	3				
F16	D3	2				
F17	C/U3	7				
F18	C/D3	26				
F19	CL	23				
F20	L	21				

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NOTE: R1 = 750Ω ±1%, 1/8 W.

FIGURE 10. Input clamp diode voltage test circuit for table VIII.

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*TABLE IX. Output voltage (low-level and high-level) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
G1	QA1 Output voltage, low level	2,3,5,8,9, 13,14,15,16, 21,23,24,26, 32,35,39,40	7,20,31	38	Open	0	400	mV	11
G2	QB1			37					
G3	QC1			18					
G4	QD1			19					
G5	QA2			33					
G6	QB2			34					
G7	QC2			30					
G8	QD2			29					
G9	QA3			27					
G10	QB3			28					
G11	QC3			6					
G12	QD3			25					
G13	BOR1			36					
G14	BOR2			12					
G15	BOR3			4					

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*TABLE IX. Output voltage (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
G16	CAR1 Output voltage, low level	7,20,21,23, 31	2,3,5,8,9, 13,14,15,16, 24,26,32,35, 39,40	17	Open	0	400	mV	11
G17	CAR2	7,20,21,23, 31	2,3,5,8,9, 13,14,15,16, 24,26,32,35, 39,40	11					
G18	CAR3	7,20,21,23, 31	2,3,5,8,9, 13,14,15,16, 24,26,32,35, 39,40	22					
H1	QA1 Output voltage, high level	21,23	2,3,5,7,8,9, 13,14,15,16, 20,24,26,31, 32,35,39,40	38		2.5	4.5	V	
H2	QB1			37					
H3	QC1			18					
H4	QD1			19					
H5	QA2			33					
H6	QB2			34					
H7	QC2			30					
H8	QD2			29					
H9	QA3			27					
H10	QB3			28					

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*TABLE IX. Output voltage (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
H11	QC3 Output voltage, high level	21,23	2,3,5,7,8,9,13,14,15,16,20,24,26,31,32,35,39,40	6	Open	2.5	4.5	V	11
H12	QD3			25					
H13	BOR1			36					
H14	BOR2			12					
H15	BOR3			4					
H16	CAR1			17					
H17	CAR2			11					
H18	CAR3			22					

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*TABLE X. Functional test

Test condition	Parameter	Connect				Output logic states																Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	QA1	QB1	QC1	QD1	CAR1	BOR1	QA2	QB2	QC2	QD2	CAR2	BOR2	QA3	QB3	QC3	QD3	CAR3	BOR3	Min	Max		
J1	Functional, parallel load	21, 23	2,3,5,7,8,9, 13,14,15,16, 20,24,26,31, 32,35,39,40	2/, 3/	Open	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H		1/		11
J2	Functional, clear	21	2,3,5,7,8,9, 13,14,15,16, 20,23,24,26, 31,32,35,39, 40		Open	L	L	L	L	H	H	L	L	L	L	H	H	L	L	L	L	H	H				
J3	Functional, parallel load	21, 23	2,3,5,8,9, 13,14,15,16, 24,26,32,39, 40		7, 20, 31	H	H	H	H	L	H	H	H	H	H	L	H	H	H	H	H	L	H				
J4		3,5,9,13, 14,15,21, 23,24,35, 40	2,7,8,16,20, 31		26,32,39	L	L	L	H	H	H	L	L	L	H	H	H	L	L	L	H	H	H				
J5		2,8,16, 21,23	3,5,9,13,14, 15,24,26,32, 35,39,40		7,20,31	H	H	H	L	H	H	H	H	H	L	H	H	H	H	H	L	H	H				
J6		3,9,15, 21,23,24, 35,40	2,5,7,8,13, 14,16,20,31		26,32,39	H	L	L	H	H	H	H	L	L	H	H	H	H	L	L	H	H	H				
J7		2,5,8,13, 14,16,21, 23	3,9,15,24, 26,32,35,39, 40		7,20,31	L	H	H	L	H	H	L	H	H	L	H	H	L	H	H	L	H	H				
J8		21,23,24, 35,40	2,3,5,7,8,9, 13,14,15,16, 20,31		26,32,39	H	L	H	H	H	H	H	L	H	H	H	H	H	L	H	H	H	H				
J9		2,3,5,8, 9,13,14, 15,16,21 ,23	24,26,32,35, 39,40		7,20,31	L	H	L	L	H	H	L	H	L	L	H	H	L	H	L	L	H	H				

See footnotes at end of table.

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*TABLE X. Functional test - Continued

Test condition	Parameter	Connect				Output logic states																		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	QA1	QB1	QC1	QD1	CAR1	BOR1	QA2	QB2	QC2	QD2	CAR2	BOR2	QA3	QB3	QC3	QD3	CAR3	BOR3	Min	Max		
J10	Functional, parallel load	5,13,14,21,23,24,35,40	2,3,7,6,9,15,16,20,31	2/, 3/	26,32,39	L	L	H	H	H	H	L	L	H	H	H	H	L	L	H	H	H	H	1/			11
J11		2,3,8,9,15,16,21,23,24,35,40	5,13,14,26,32,39		7,20,31	H	L	L	L	H	H	H	L	L	L	H	H	H	L	L	L	H	H				
J12		5,13,14,21,23	2,3,7,8,9,15,16,20,24,31,35,40		26,32,39	L	H	H	H	H	H	L	H	H	H	H	H	L	H	H	H	H	H				
J13		2,3,8,9,15,16,21,23	5,13,14,24,26,32,35,39,40		7,20,31	H	H	L	L	H	H	H	H	L	L	H	H	H	H	L	L	H	H				
J14		2,5,8,13,14,16,21,23,24,35,40	3,7,9,15,20,31		26,32,39	L	L	H	L	H	H	L	L	H	L	H	H	L	L	H	L	H	H				
J15		3,9,15,21,23	2,5,8,13,14,16,24,26,32,35,39,40		7,20,31	H	H	L	H	H	H	H	H	L	H	H	H	H	H	L	H	H	H				
J16		2,8,16,21,23,24,35,40	3,5,7,9,13,14,15,20,31		26,32,39	H	L	H	L	H	H	H	L	H	L	H	H	H	L	H	L	H	H				
J17		3,5,9,13,14,15,21,23	2,8,16,24,26,32,35,39,40		7,20,31	L	H	L	H	H	H	L	H	L	H	H	H	L	H	L	H	H	H				
J18		7,20,21,23,31	2,3,5,8,9,13,14,15,16,24,26,32,35,39,40		Open	H	H	H	H	L	H	H	H	H	H	L	H	H	H	H	H	L	H				
J19		2,3,5,8,9,13,14,15,16,21,23,24,26,32,35,39,40	7,20,31		Open	L	L	L	L	H	L	L	L	L	L	H	L	L	L	L	L	H	L				

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*TABLE X. Functional test - Continued.

Test condition	Parameter	Connect				Output logic states																	Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	QA1	QB1	QC1	QD1	CAR1	BOR1	QA2	QB2	QC2	QD2	CAR2	BOR2	QA3	QB3	QC3	QD3	CAR3	BOR3				
																								Min	Max		
J20	Functional, parallel load	2,3,5,8, 9,13,14, 15,16,21, 23,24,35, 40	7,20,31	2/, 3/	26,32,39	L	L	L	L	H	L	L	L	L	L	H	L	L	L	L	L	H	L		1/		11
J21	Functional, count up	2,3,5,8, 9,13,14, 15,16,23, 24,35,40	21,26,32,39		7,20,31	H	L	L	L	H	H	H	L	L	L	H	H	H	L	L	L	H	H				
J22						L	H	L	L	H	H	L	H	L	L	H	H	L	H	L	L	H	H				
J23						H	H	L	L	H	H	H	H	L	L	H	H	H	H	L	L	H	H				
J24						L	L	H	L	H	H	L	L	H	L	H	H	L	L	H	L	H	H				
J25						H	L	H	L	H	H	H	L	H	L	H	H	H	L	H	L	H	H				
J26						L	H	H	L	H	H	L	H	H	L	H	H	L	H	H	L	H	H				
J27						H	H	H	L	H	H	H	H	H	L	H	H	H	H	H	L	H	H				
J28						L	L	L	H	H	H	L	L	L	H	H	H	L	L	L	H	H	H				
J29						H	L	L	H	H	H	H	L	L	H	H	H	H	L	L	H	H	H				
J30						L	H	L	H	H	H	L	H	L	H	H	H	L	H	L	H	H	H				
J31						H	H	L	H	H	H	H	H	L	H	H	H	H	H	L	H	H	H				
J32						L	L	H	H	H	H	L	L	H	H	H	H	L	L	H	H	H	H				
J33						H	L	H	H	H	H	H	L	H	H	H	H	H	L	H	H	H	H				
J34						L	H	H	H	H	H	L	H	H	H	H	H	L	H	H	H	H	H				
J35						H	H	H	H	L	H	H	H	H	H	L	H	H	H	H	H	L	H				
J36		23	2,3,5,8,9, 13,14,15,16, 21,24,26,32, 35,39,40			L	L	L	L	H	H	L	L	L	L	H	H	L	L	L	L	H	H				

See footnotes at end of table.

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*TABLE X. Functional test - Continued.

Test condition	Parameter	Connect				Output logic states																		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	QA1	QB1	QC1	QD1	CAR1	BOR1	QA2	QB2	QC2	QD2	CAR2	BOR2	QA3	QB3	QC3	QD3	CAR3	BOR3	Min	Max		
J37	Functional, parallel load	21,23	2,3,5,7,8,9, 13,14,15,16, 20,24,31,35, 40	2/, 3/	26,32,39	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H		1/		11
J38	Functional, count down	23				L	H	H	H	H	H	L	H	H	H	H	H	L	H	H	H	H	H				
J39						H	L	H	H	H	H	H	L	H	H	H	H	H	L	H	H	H	H				
J40						L	L	H	H	H	H	L	L	H	H	H	H	L	L	H	H	H	H				
J41						H	H	L	H	H	H	H	H	L	H	H	H	H	H	L	H	H	H				
J42						L	H	L	H	H	H	L	H	L	H	H	H	L	H	L	H	H	H				
J43						H	L	L	H	H	H	H	L	L	H	H	H	H	L	L	H	H	H				
J44						L	L	L	H	H	H	L	L	L	H	H	H	L	L	L	H	H	H				
J45						H	H	H	L	H	H	H	H	H	L	H	H	H	H	H	L	H	H				
J46						L	H	H	L	H	H	L	H	H	L	H	H	L	H	H	L	H	H				
J47						H	L	H	L	H	H	H	L	H	L	H	H	H	L	H	L	H	H				
J48						L	L	H	L	H	H	L	L	H	L	H	H	L	L	H	L	H	H				
J49						H	H	L	L	H	H	H	H	L	L	H	H	H	H	L	L	H	H				
J50						L	H	L	L	H	H	L	H	L	L	H	H	L	H	L	L	H	H				
J51						H	L	L	L	H	H	H	L	L	L	H	H	H	L	L	L	H	H				
J52						L	L	L	L	H	L	L	L	L	L	H	L	L	L	L	L	H	L				
J53						H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H				

1/ Output logic states:
L = Low-level = 0.0 to 0.4 V.
H = High-level = 2.5 to 4.5 V.

2/ For test conditions J1-J53, vertification of the output logic states in footnote 1 is obtained by connecting item M02 (point C) to each output pin in succession after the input conditions have been accomplished. Test conditions J1-J53 are to be performed in sequence.

3/ QA1, QB1, QC1, QD1, CAR1, BOR1, QA2, QB2, QC2, QD2, CAR2, BOR2, QA3, QB3, QC3, QD3, CAR3, and BOR3 output logic states are measured at output pins 38, 37, 18, 19, 17, 36, 33, 34, 30, 29, 11, 12, 27, 28, 6, 25, 22, and 4 respectively.

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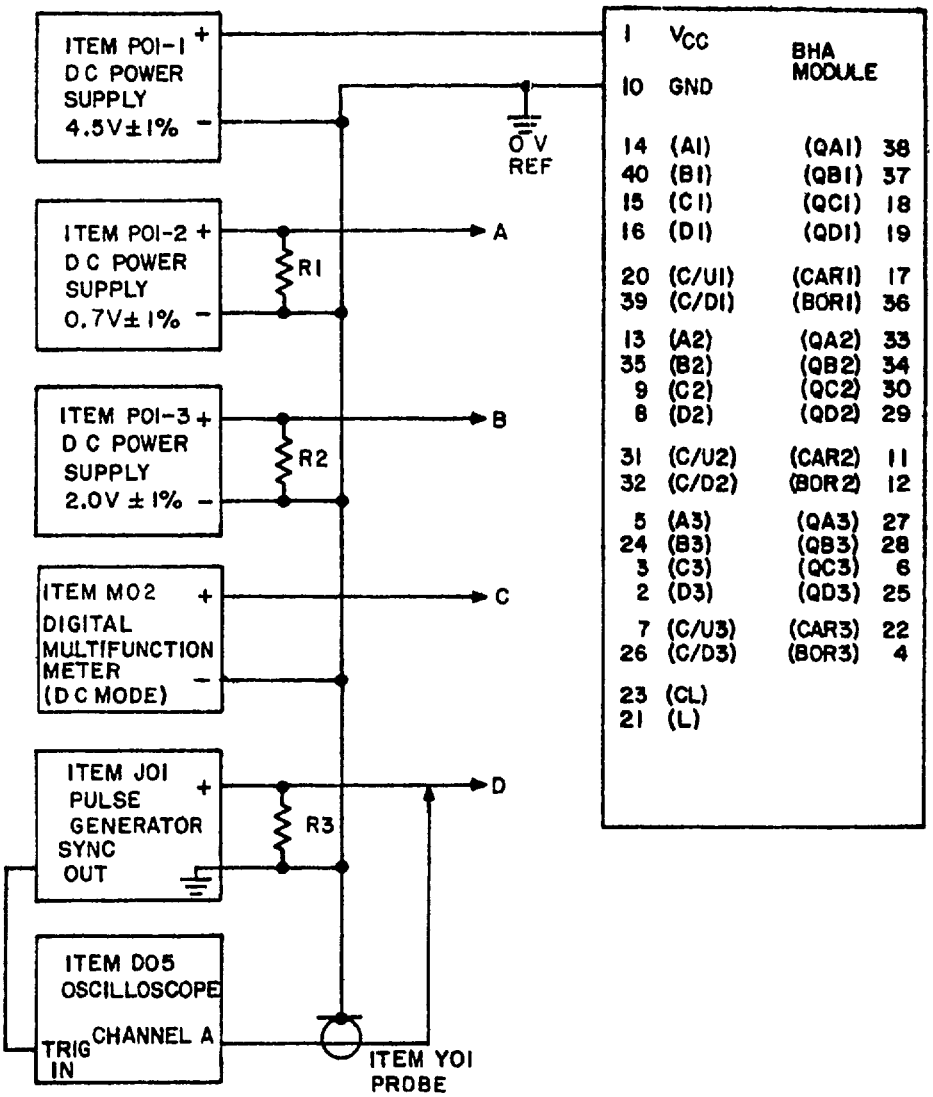


FIGURE 11. Output voltage (low-level and high-level) and functional test circuit for tables IX and X.

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NOTES:
1. R1-R3 = 51Ω ±5%, 1/2 W.
2. For test conditions G1-G18 and H1-H18, load each output with load as specified on figure 12
3. For test conditions J1-J53, loading of outputs is not required.
4. For test conditions J1-J53:
   After making all input circuit connections for a test condition, use the single pulse mode on item J01 (after initial setup) and apply one pulse before making each measurement. Test conditions J1-J53 are to be performed in sequence.
5. Item J01 setup.
   PW = 500 ±50 ns.
   $t_{TLH} = t_{THL}$ = 50 ±25 ns.
   Place item J01 in the single pulse mode.

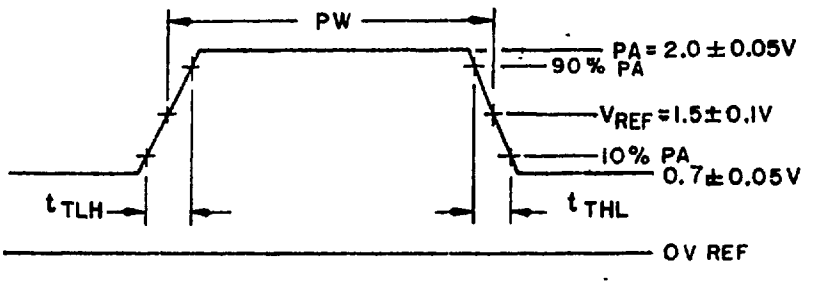
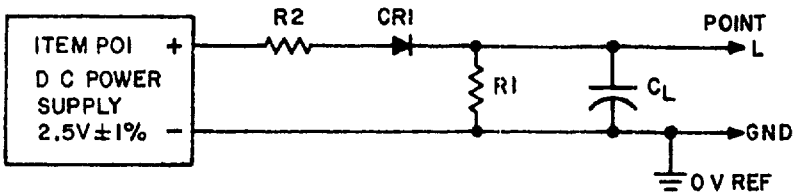


FIGURE 11. Output voltage (low-level and high-level) and functional test circuit for tables IX and X - Continued.

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NOTES:
1. CR1 = 1N4150, or equivalent.
2. R1 = 6.19 kΩ ±1%, 1/8 W.
* 3. R2 = select value for a current flow of 4.0 mA ±5% out of point L with point L held at 0.4 V.
4 C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions N1-N18, P1-P18, Q1-Q6, R1-R6, S1-S12, T1-T12, U1-U12, V1-V12, W1-W12, X1-X12, Y1-Y12, Z1-Z6, AA1-AA12, BB1-BB12 and CC1-CC24.
5. C_L = 150 ±15 pF (including probe and parasitic capacitance) for test conditions N19-N36, P19-P36, Q7-Q12, R7-R12, S13-S24, T13-T24, U13-U24, V13-V24, W13-W24, X13-X24 and Y13-Y24.
6. C_L is not required for test conditions G, H and J.

FIGURE 12. Logic load circuit.

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TABLE XI. Input current (low-level and high-level) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
K1	A1 Input current, low level	21, 23	Open	14	Open	-0.010	-0.435	mA	13
K2	B1			40					
K3	C1			15					
K4	D1			16					
K5	C/U1	Open		20					
K6	C/D1	Open		39					
K7	A2	21, 23		13					
K8	B2			35					
K9	C2			9					
K10	D2			8					
K11	C/U2	Open		31					
K12	C/D2	Open		32					
K13	A3	21, 23		5					
K14	B3			24					
K15	C3			3					
K16	D3			2					
K17	C/U3	Open		7					
K18	C/D3			26					
K19	CL			23		-0.03	-1.30		
K20	L			21		-0.03	-1.30		

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TABLE XI. Input current (low-level and high-level) test - Continued.

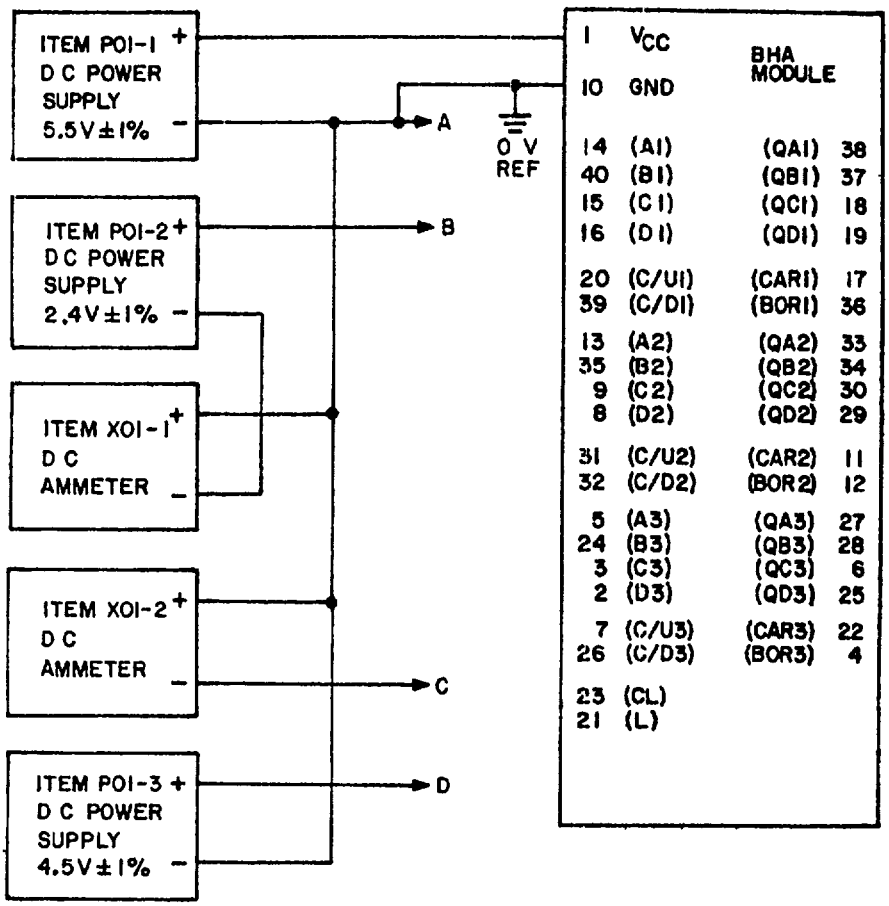
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
L1	A1 Input current, high level	15,16,20,39,40	14	Open	21, 23	0	20	µA	13
L2	B1	14,15,16,20,39	40						
L3	C1	14,16,20,39,40	15						
L4	D1	14,15,20,39,40	16						
L5	C/U1	14,15,16,21,23,39,40	20		Open				
L6	C/D1	14,15,16,20,21,23,40	39		Open				
L7	A2	8,9,31,32,35	13		21, 23				
L8	B2	8,9,13,31,32	35						
L9	C2	8,13,31,32,35	9						
L10	D2	9,13,31,32,35	8						
L11	C/U2	8,9,13,21,23,32,35	31		Open				
L12	C/D2	8,9,13,21,23,31,35	32		Open				
L13	A3	2,3,7,24,26	5		21, 23				

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TABLE XI. Input current (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
L14	B3 Input current, high level	2,3,5,7,26	24	Open	21, 23	0	20	µA	13
L15	C3	2,5,7,24,26	3		21, 23				
L16	D3	3,5,7,24,26	2		21, 23				
L17	C/U3	2,3,5,21,23, 24,26	7		Open				
L18	C/D3	2,3,5,7,21, 23,24	26						
L19	CL	2,3,5,7,8,9, 13,14,15,16, 20,21,24,26, 31,32,35,39, 40	23				60		
L20	L	2,3,5,7,8,9, 13,14,15,16, 20,23,24,26, 31,32,35,39, 40	21				60		

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CAUTION

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. Since items X01-1 and X01-2 are not used simultaneously, only one item X01 is required.
2. Only one output shall be shorted at a time, and the measurement taken within 10 seconds.

FIGURE 13. Input current (low-level and high-level) test circuit for table XI.

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TABLE XII. Power supply current (high-level and low-level) test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient		Unit	Test figure
			Min	Max		
M1	Power supply current, high level	21, 23	10	102	mA	14
M2	Power supply current, low level	Open	10	102	mA	14

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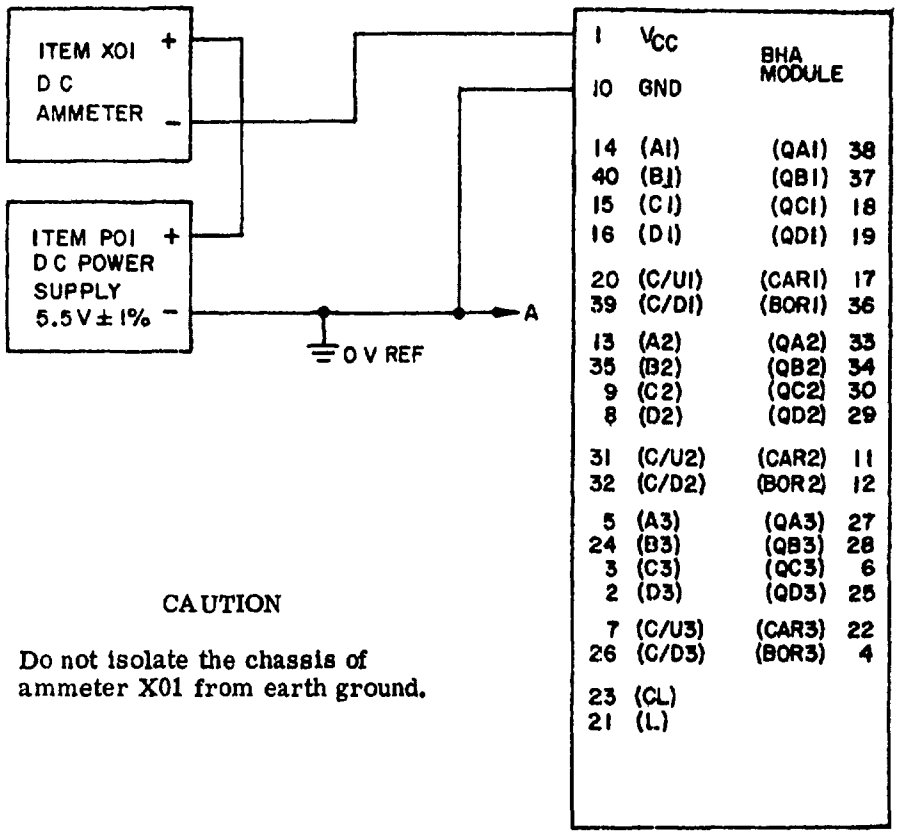


FIGURE 14. Power supply current (high-level and low-level) test circuit for table XII.

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*TABLE XIII. Transition time (t_{TLH} and t_{THL}) test.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
N1	CAR1 Transition time (t_{TLH}), C_L = 50 pF	14,15,16, 40	21,26, 32,39	20	Open	20	17	5	50	ns	15,16
N2	CAR2	8,9,13, 35	21,26, 32,39	31		31	11				
N3	CAR3	2,3,5, 24	21,26, 32,39	7		7	22				
N4	BOR1	14,15, 16,40	7,20, 21,31	39		39	36				
N5	BOR2	8,9,13, 35	7,20, 21,31	32		32	12				
N6	BOR3	2,3,5, 24	7,20, 21,31	26		26	4				
N7	QA1	14,15, 16,40	Open	20	21	20	38				15, 17
N8	QA2	8,9,13, 35	Open	31		31	33				
N9	QA3	2,3,5, 24	Open	7		7	27				
N10	QB1	15,16,40	14	20		20	37				
N11	QB2	8,9,35	13	31		31	34				
N12	QB3	2,3,24	5	7		7	28				
N13	QC1	15,16	14,40	20		20	18				
N14	QC2	8,9	13,35	31		31	30				

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*TABLE XIII. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter	Connect: A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Initial limits 25°±5°C ambient: Min	Max	Unit	Test figure
N15	QC3 Transition time (t_{TLH}), C_L = 50 pF	2,3	5,24	7	21	7	6	5	50	ns	15,17
N16	QD1	16	14,15, 40	20		20	19				
N17	QD2	8	9,13,35	31		31	29				
N18	QD3	2	3,5,24	7		7	25				
N19	CAR1 Transition time (t_{TLH}), C_L = 150 pF	14	15,16, 40	20		20	17	10	80		15,16
N20	CAR2	13	9,8,35	31		31	11				
N21	CAR3	5	3,2,24	7		7	22				
N22	BOR1	15,16,40	14	39		39	36				
N23	BOR2	8,9,35	13	32		32	12				
N24	BOR3	2,3,24	5	26		26	4				
N25	QA1	14,15,16, 40	Open	20		20	38				15,17
N26	QA2	8,9,13,35	Open	31		31	33				
N27	QA3	2,3,5,24	Open	7		7	27				
N28	QB1	15,16,40	14	20		20	37				
N29	QB2	8,9,35	13	31		31	34				
N30	QB3	2,3,24	5	7		7	28				
N31	QC1	15,16	14,40	20		20	18				
N32	QC2	8,9	13,35	31		31	30				

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*TABLE XIII. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Connect F to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
N33	QC3 Transition time (t_{TLH}), C_L = 150 pF	2,3	5,24	7	21	7	6	10	80	ns	15,17
N34	QD1	16	14,15, 40	20		20	19				
N35	QD2	8	9,13,35	31		31	29				
N36	QD3	2	3,5	7		7	25				
P1	CAR1 Transition time (t_{THL}), C_L = 50 pF	14,15, 16,40	21,26, 32,39	20	Open	20	17	3	18		15,16
P2	CAR2	8,9,13, 35	21,26, 32,39	31		31	11				
P3	CAR3	2,3,5, 24	21,26, 32,39	7		7	22				
P4	BOR1	14,15, 16,40	7,20, 21,31	39		39	36				
P5	BOR2	8,9,13, 35	7,20, 21,31	32		32	12				
P6	BOR3	2,3,5, 24	7,20, 21,31	26		26	4				
P7	QA1	15,16,40	14	20	21	20	38				15, 17
P8	QA2	8,9,35	13	31		31	33				
P9	QA3	2,3,24	5	7		7	27				
P10	QB1	15,16	14,40	20		20	37				
P11	QB2	8,9	13,35	31		31	34				
P12	QB3	2,3	5,24	7		7	28				
P13	QC1	16	14,15, 40	20		20	18				
P14	QC2	8	9,13,35	31		31	30				

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*TABLE XIII. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
P15	QC3 Transition time (t_{THL}), C_L = 50 pF	2	3,5,24	7	21	7	6	3	18	ns	15,17
P16	QD1	Open	14,16, 15,40	20		20	19				
P17	QD2	Open	8,13, 9,35	31		31	29				
P18	QD3	Open	2,5, 3,24	7		7	25				
P19	CAR1 Transition time (t_{THL}), C_L = 150 pF	14	15,16, 40	20		20	17	6	40		15,16
P20	CAR2	13	8,9,35	31		31	11				
P21	CAR3	5	2,3,24	7		7	22				
P22	BOR1	15,16,40	14	39		39	36				
P23	BOR2	8,9,35	13	32		32	12				
P24	BOR3	2,3,24	5	26		26	4				
P25	QA1	15,16,40	14	20		20	38				15,17
P26	QA2	8,9,35	13	31		31	33				
P27	QA3	2,3,24	5	7		7	27				
P28	QB1	15,16	14,40	20		20	37				
P29	QB2	8,9	13,35	31		31	34				
P30	QB3	2,3	5,24	7		7	28				
P31	QC1	16	14,15, 40	20		20	18				
P32	QC2	8	9,13,35	31		31	30				

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*TABLE XIII. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
P33	QC3 Transition time (t_{THL}), C_L = 150 pF	2	3,5,24	7	21	7	6	6	40	ns	15,17
P34	QD1	Open	14,16, 15,40	20		20	19				
P35	QD2	Open	8,13, 9,35	31		31	29				
P36	QD3	Open	2,5, 3,24	7		7	25				

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*TABLE XIV. Propagation delay time (t_{PHL} and t_{PLH}) test.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
Q1	Propagation delay time (t_{PHL}), count up to CAR1, C_L = 50 pF	14,15, 16,40	21,26, 32,39	20	Open	20	17	3	28	ns	15,16
Q2	Propagation delay time (t_{PHL}), count up to CAR2, C_L = 50 pF	8,9, 13,35	21,26, 32,39	31		31	11				
Q3	Propagation delay time (t_{PHL}), count up to CAR3, C_L = 50 pF	2,3, 5,24	21,26, 32,39	7		7	22				
Q4	Propagation delay time (t_{PHL}), count down to BOR1, C_L = 50 pF	14,15, 16,40	7,20, 21,31	39		39	36				
Q5	Propagation delay time (t_{PHL}), count down to BOR2, C_L = 50 pF	8,9, 13,35	7,20, 21,31	32		32	12				
Q6	Propagation delay time (t_{PHL}), count down to BOR3, C_L = 50 pF	2,3, 5,24	7,20, 21,31	26		26	4				
Q7	Propagation delay time (t_{PHL}), count up to CAR1, C_L = 150 pF	14	15,16, 40	20	21	20	17	4	38		
Q8	Propagation delay time (t_{PHL}), count up to CAR2, C_L = 150 pF	13	9,8, 35	31		31	11				
Q9	Propagation delay time (t_{PHL}), count up to CAR3, C_L = 150 pF	5	3,2, 24	7		7	22				
Q10	Propagation delay time (t_{PHL}), count down to BOR1, C_L = 150 pF	15,16 40	14	39		39	36				
Q11	Propagation delay time (t_{PHL}), count down to BOR2, C_L = 150 pF	8,9, 35	13	32		32	12				
Q12	Propagation delay time (t_{PHL}), count down to BOR3, C_L = 150 pF	2,3, 24	5	26		26	4				

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*TABLE XIV. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
R1	Propagation delay time (t_{PLH}), count up to CAR1, C_L = 50 pF	14,15, 16,40	21,26, 32,39	20	Open	20	17	3	30	ns	15,16
R2	Propagation delay time (t_{PLH}), count up to CAR2, C_L = 50 pF	8,9, 13,35	21,26, 32,39	31		31	11		30		
R3	Propagation delay time (t_{PLH}), count up to CAR3, C_L = 50 pF	2,3, 5,24	21,26, 32,39	7		7	22		30		
R4	Propagation delay time (t_{PLH}), count down to BOR1, C_L = 50 pF	14,15, 16,40	7,20, 21,31	39		39	36		28		
R5	Propagation delay time (t_{PLH}), count down to BOR2, C_L = 50 pF	8,9, 13,35	7,20, 21,31	32		32	12		28		
R6	Propagation delay time (t_{PLH}), count down to BOR3, C_L = 50 pF	2,3, 5,24	7,20, 21,31	26		26	4		28		
R7	Propagation delay time (t_{PLH}), count up to CAR1, C_L = 150 pF	14	15,16, 40	20	21	20	17	4	40		
R8	Propagation delay time (t_{PLH}), count up to CAR2, C_L = 150 pF	13	9,8, 35	31		31	11		40		
R9	Propagation delay time (t_{PLH}), count up to CAR3, C_L = 150 pF	5	3,2, 24	7		7	22		40		
R10	Propagation delay time (t_{PLH}), count down to BOR1, C_L = 150 pF	15,16 40	14	39		39	36		38		
R11	Propagation delay time (t_{PLH}), count down to BOR2, C_L = 150 pF	8,9, 35	13	32		32	12		38		
R12	Propagation delay time (t_{PLH}), count down to BOR3, C_L = 150 pF	2,3, 24	5	26		26	4		38		

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*TABLE XV. Propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
S1	Propagation delay time (t_{PLH}), count up to QA1, C_L = 50 pF	14,15, 16,40	Open	20	21	20	38	4	42	ns	15,17
S2	Propagation delay time (t_{PLH}), count up to QA2, C_L = 50 pF	8,9, 13,35	Open	31		31	33				
S3	Propagation delay time (t_{PLH}), count up to QA3, C_L = 50 pF	2,3, 5,24	Open	7		7	27				
S4	Propagation delay time (t_{PLH}), count up to QB1, C_L = 50 pF	15,16, 40	14	20		20	37				
S5	Propagation delay time (t_{PLH}), count up to QB2, C_L = 50 pF	8,9, 35	13	31		31	34				
S6	Propagation delay time (t_{PLH}), count up to QB3, C_L = 50 pF	2,3, 24	5	7		7	28				
S7	Propagation delay time (t_{PLH}), count up to QC1, C_L = 50 pF	15,16	14,40	20		20	18				
S8	Propagation delay time (t_{PLH}), count up to QC2, C_L = 50 pF	8,9	13,35	31		31	30				
S9	Propagation delay time (t_{PLH}), count up to QC3, C_L = 50 pF	2,3	5,24	7		7	6				
S10	Propagation delay time (t_{PLH}), count up to QD1, C_L = 50 pF	16	14,15, 40	20		20	19				
S11	Propagation delay time (t_{PLH}), count up to QD2, C_L = 50 pF	8	9,13, 35	31		31	29				
S12	Propagation delay time (t_{PLH}), count up to QD3, C_L = 50 pF	2	3,5, 24	7		7	25				

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*TABLE XV. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
S13	Propagation delay time (t_{PLH}), count up to QA1, C_L = 150 pF	14,15, 16,40	Open	20	21	20	38	5	52	ns	15,17
S14	Propagation delay time (t_{PLH}), count up to QA2, C_L = 150 pF	8,9, 13,35	Open	31		31	33				
S15	Propagation delay time (t_{PLH}), count up to QA3, C_L = 150 pF	2,3, 5,24	Open	7		7	27				
S16	Propagation delay time (t_{PLH}), count up to QB1, C_L = 150 pF	15,16, 40	14	20		20	37				
S17	Propagation delay time (t_{PLH}), count up to QB2, C_L = 150 pF	8,9, 35	13	31		31	34				
S18	Propagation delay time (t_{PLH}), count up to QB3, C_L = 150 pF	2,3, 24	5	7		7	28				
S19	Propagation delay time (t_{PLH}), count up to QC1, C_L = 150 pF	15,16	14,40	20		20	18				
S20	Propagation delay time (t_{PLH}), count up to QC2, C_L = 150 pF	8,9	13,35	31		31	30				
S21	Propagation delay time (t_{PLH}), count up to QC3, C_L = 150 pF	2,3	5,24	7		7	6				
S22	Propagation delay time (t_{PLH}), count up to QD1, C_L = 150 pF	16	14,15, 40	20		20	19				
S23	Propagation delay time (t_{PLH}), count up to QD2, C_L = 150 pF	8	9,13, 35	31		31	29				
S24	Propagation delay time (t_{PLH}), count up to QD3, C_L = 150 pF	2	3,5, 24	7		7	25				

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*TABLE XV. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
T1	Propagation delay time (t_{PHL}), count up to QA1, C_L = 50 pF	15,16, 40	14	20	21	20	38	5	51	ns	15,17
T2	Propagation delay time (t_{PHL}), count up to QA2, C_L = 50 pF	8,9, 35	13	31		31	33				
T3	Propagation delay time (t_{PHL}), count up to QA3, C_L = 50 pF	2,3, 24	5	7		7	27				
T4	Propagation delay time (t_{PHL}), count up to QB1, C_L = 50 pF	15,16	14,40	20		20	37				
T5	Propagation delay time (t_{PHL}), count up to QB2, C_L = 50 pF	8,9	13,35	31		31	34				
T6	Propagation delay time (t_{PHL}), count up to QB3, C_L = 50 pF	2,3	5,24	7		7	28				
T7	Propagation delay time (t_{PHL}), count up to QC1, C_L = 50 pF	16	14,15, 40	20		20	18				
T8	Propagation delay time (t_{PHL}), count up to QC2, C_L = 50 pF	8	9,13, 35	31		31	30				
T9	Propagation delay time (t_{PHL}), count up to QC3, C_L = 50 pF	2	3,5, 24	7		7	6				
T10	Propagation delay time (t_{PHL}), count up to QD1, C_L = 50 pF	Open	14,16, 15,40	20		20	19				
T11	Propagation delay time (t_{PHL}), count up to QD2, C_L = 50 pF	Open	8,13, 9,35	31		31	29				
T12	Propagation delay time (t_{PHL}), count up to QD3, C_L = 50 pF	Open	2,5,3, 24	7		7	25				

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*TABLE XV. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
T13	Propagation delay time (t_{PHL}), count up to QA1, C_L = 150 pF	15,16, 40	14	20	21	20	38	6	61	ns	15,17
T14	Propagation delay time (t_{PHL}), count up to QA2, C_L = 150 pF	8,9, 35	13	31		31	33				
T15	Propagation delay time (t_{PHL}), count up to QA3, C_L = 150 pF	2,3, 24	5	7		7	27				
T16	Propagation delay time (t_{PHL}), count up to QB1, C_L = 150 pF	15,16,	14,40	20		20	37				
T17	Propagation delay time (t_{PHL}), count up to QB2, C_L = 150 pF	8,9,	13,35	31		31	34				
T18	Propagation delay time (t_{PHL}), count up to QB3, C_L = 150 pF	2,3,	5,24	7		7	28				
T19	Propagation delay time (t_{PHL}), count up to QC1, C_L = 150 pF	16	14,15, 40	20		20	18				
T20	Propagation delay time (t_{PHL}), count up to QC2, C_L = 150 pF	8	9,13, 35	31		31	30				
T21	Propagation delay time (t_{PHL}), count up to QC3, C_L = 150 pF	2	3,5, 24	7		7	6				
T22	Propagation delay time (t_{PHL}), count up to QD1, C_L = 150 pF	Open	14,16, 15,40	20		20	19				
T23	Propagation delay time (t_{PHL}), count up to QD2, C_L = 150 pF	Open	8,13, 9,35	31		31	29				
T24	Propagation delay time (t_{PHL}), count up to QD3, C_L = 150 pF	Open	2,5 3,24	7		7	25				

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*TABLE XV. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
U1	Propagation delay time (t_{PLH}), count down to QA1, C_L = 50 pF	14,15, 16	40	39	21	39	38	4	42	ns	15,17
U2	Propagation delay time (t_{PLH}), count down to QA2, C_L = 50 pF	8,9, 13	35	32	↓	32	33	↓	↓	↓	↓
U3	Propagation delay time (t_{PLH}), count down to QA3, C_L = 50 pF	2,3, 5	24	26	↓	26	27	↓	↓	↓	↓
U4	Propagation delay time (t_{PLH}), count down to QB1, C_L = 50 pF	14,16, 40	15	39	↓	39	37	↓	↓	↓	↓
U5	Propagation delay time (t_{PLH}), count down to QB2, C_L = 50 pF	8,13, 35	9	32	↓	32	34	↓	↓	↓	↓
U6	Propagation delay time (t_{PLH}), count down to QB3, C_L = 50 pF	2,5, 24	3	26	↓	26	28	↓	↓	↓	↓
U7	Propagation delay time (t_{PLH}), count down to QC1, C_L = 50 pF	14,15, 40	16	39	↓	39	18	↓	↓	↓	↓
U8	Propagation delay time (t_{PLH}), count down to QC2, C_L = 50 pF	9,13, 35	8	32	↓	32	30	↓	↓	↓	↓
U9	Propagation delay time (t_{PLH}), count down to QC3, C_L = 50 pF	3,5, 24	2	26	↓	26	6	↓	↓	↓	↓
U10	Propagation delay time (t_{PLH}), count down to QD1, C_L = 50 pF	14,15, 16,40	Open	39	↓	39	19	↓	↓	↓	↓
U11	Propagation delay time (t_{PLH}), count down to QD2, C_L = 50 pF	8,9, 13,35	Open	32	↓	32	29	↓	↓	↓	↓
U12	Propagation delay time (t_{PLH}), count down to QD3, C_L = 50 pF	2,3, 5,24	Open	26	↓	26	25	↓	↓	↓	↓

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*TABLE XV. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
U13	Propagation delay time (t_{PLH}), count down to QA1, C_L = 150 pF	14,15, 16	40	39	21	39	38	5	52	ns	15,17
U14	Propagation delay time (t_{PLH}), count down to QA2, C_L = 150 pF	8,9, 13	35	32		32	33				
U15	Propagation delay time (t_{PLH}), count down to QA3, C_L = 150 pF	2,3, 5	24	26		26	27				
U16	Propagation delay time (t_{PLH}), count down to QB1, C_L = 150 pF	14,16, 40	15	39		39	37				
U17	Propagation delay time (t_{PLH}), count down to QB2, C_L = 150 pF	8,13, 35	9	32		32	34				
U18	Propagation delay time (t_{PLH}), count down to QB3, C_L = 150 pF	2,5, 24	3	26		26	28				
U19	Propagation delay time (t_{PLH}), count down to QC1, C_L = 150 pF	14,15, 40	16	39		39	18				
U20	Propagation delay time (t_{PLH}), count down to QC2, C_L = 150 pF	9,13, 35	8	32		32	30				
U21	Propagation delay time (t_{PLH}), count down to QC3, C_L = 150 pF	3,5, 24	2	26		26	6				
U22	Propagation delay time (t_{PLH}), count down to QD1, C_L = 150 pF	14,15, 16,40	Open	39		39	19				
U23	Propagation delay time (t_{PLH}), count down to QD2, C_L = 150 pF	8,9, 13,35	Open	32		32	29				
U24	Propagation delay time (t_{PLH}), count down to QD3, C_L = 150 pF	2,3, 5,24	Open	26		26	25				

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*TABLE XV. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued

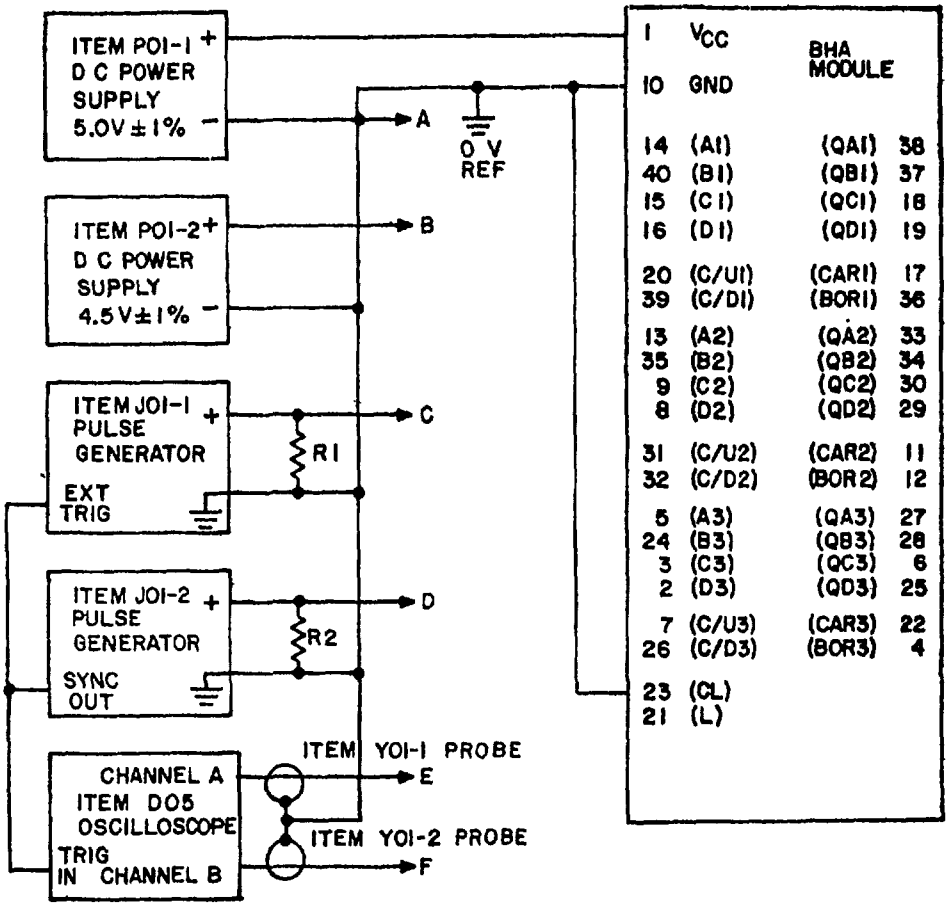
Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
V1	Propagation delay time (t_{PHL}), count down to QA1, C_L = 50 pF	15,16, 40	14	39	21	39	38	5	51	ns	15,17
V2	Propagation delay time (t_{PHL}), count down to QA2, C_L = 50 pF	8,9, 35	13	32		32	33				
V3	Propagation delay time (t_{PHL}), count down to QA3, C_L = 50 pF	2,3, 24	5	26		26	27				
V4	Propagation delay time (t_{PHL}), count down to QB1, C_L = 50 pF	14,15, 16	40	39		39	37				
V5	Propagation delay time (t_{PHL}), count down to QB2, C_L = 50 pF	8,9, 13	35	32		32	34				
V6	Propagation delay time (t_{PHL}), count down to QB3, C_L = 50 pF	2,3, 5	24	26		26	28				
V7	Propagation delay time (t_{PHL}), count down to QC1, C_L = 50 pF	14,16, 40	15	39		39	18				
V8	Propagation delay time (t_{PHL}), count down to QC2, C_L = 50 pF	8,13, 35	9	32		32	30				
V9	Propagation delay time (t_{PHL}), count down to QC3, C_L = 50 pF	2,5, 24	3	26		26	6				
V10	Propagation delay time (t_{PHL}), count down to QD1, C_L = 50 pF	14,15, 40	16	39		39	19				
V11	Propagation delay time (t_{PHL}), count down to QD2, C_L = 50 pF	9, 13,35	8	32		32	29				
V12	Propagation delay time (t_{PHL}), count down to QD3, C_L = 50 pF	3,5, 24	2	26		26	25				

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*TABLE XV. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
V13	Propagation delay time (t_{PHL}), count down to QA1, C_L = 150 pF	15,16, 40	14	39	21	39	38	6	61	ns	15,17
V14	Propagation delay time (t_{PHL}), count down to QA2, C_L = 150 pF	8,9, 35	13	32		32	33				
V15	Propagation delay time (t_{PHL}), count down to QA3, C_L = 150 pF	2,3, 24	5	26		26	27				
V16	Propagation delay time (t_{PHL}), count down to QB1, C_L = 150 pF	14,15, 16	40	39		39	37				
V17	Propagation delay time (t_{PHL}), count down to QB2, C_L = 150 pF	8,9, 13	35	32		32	34				
V18	Propagation delay time (t_{PHL}), count down to QB3, C_L = 150 pF	2,3, 5	24	26		26	28				
V19	Propagation delay time (t_{PHL}), count down to QC1, C_L = 150 pF	14,16, 40	15	39		39	18				
V20	Propagation delay time (t_{PHL}), count down to QC2, C_L = 150 pF	8,13, 35	9	32		32	30				
V21	Propagation delay time (t_{PHL}), count down to QC3, C_L = 150 pF	2,5, 24	3	26		26	6				
V22	Propagation delay time (t_{PHL}), count down to QD1, C_L = 150 pF	14,15, 40	16	39		39	19				
V23	Propagation delay time (t_{PHL}), count down to QD2, C_L = 150 pF	9,13, 35	8	32		32	29				
V24	Propagation delay time (t_{PHL}), count down to QD3, C_L = 150 pF	3,5, 24	2	26		26	25				

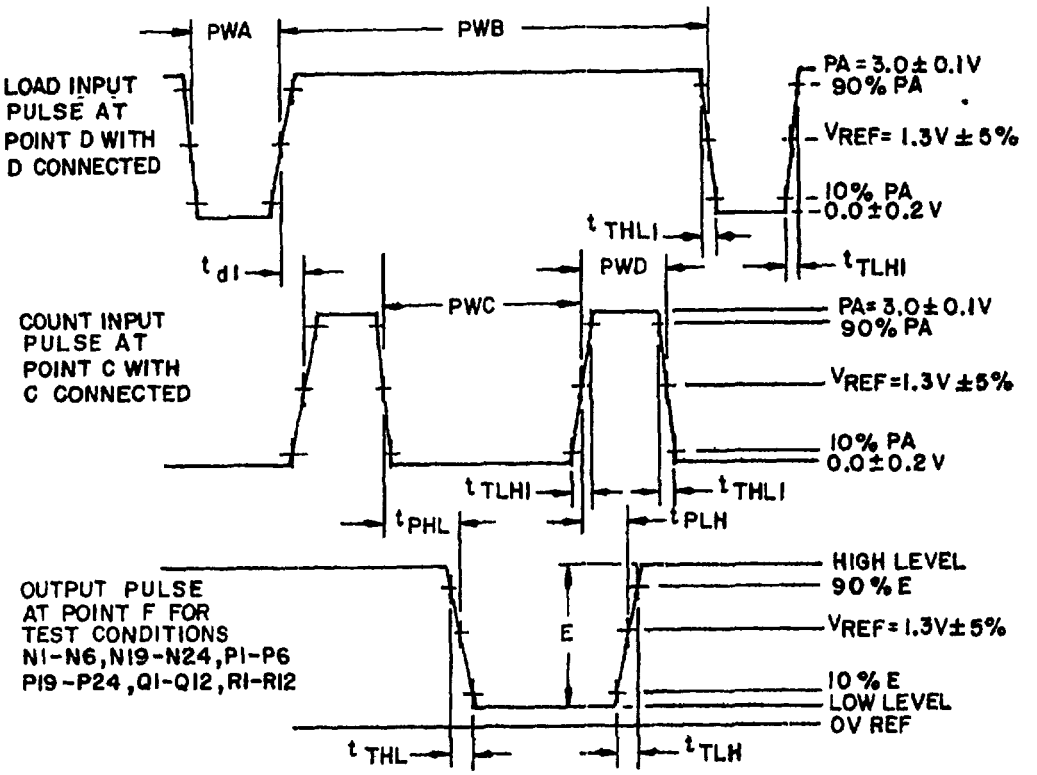
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. Load all outputs with load as specified on figure 12.

* FIGURE 15. Transition time (t_{TLH} and t_{THL}), propagation delay time (t_{PHL} and t_{PLH}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XIII, XIV and XV.

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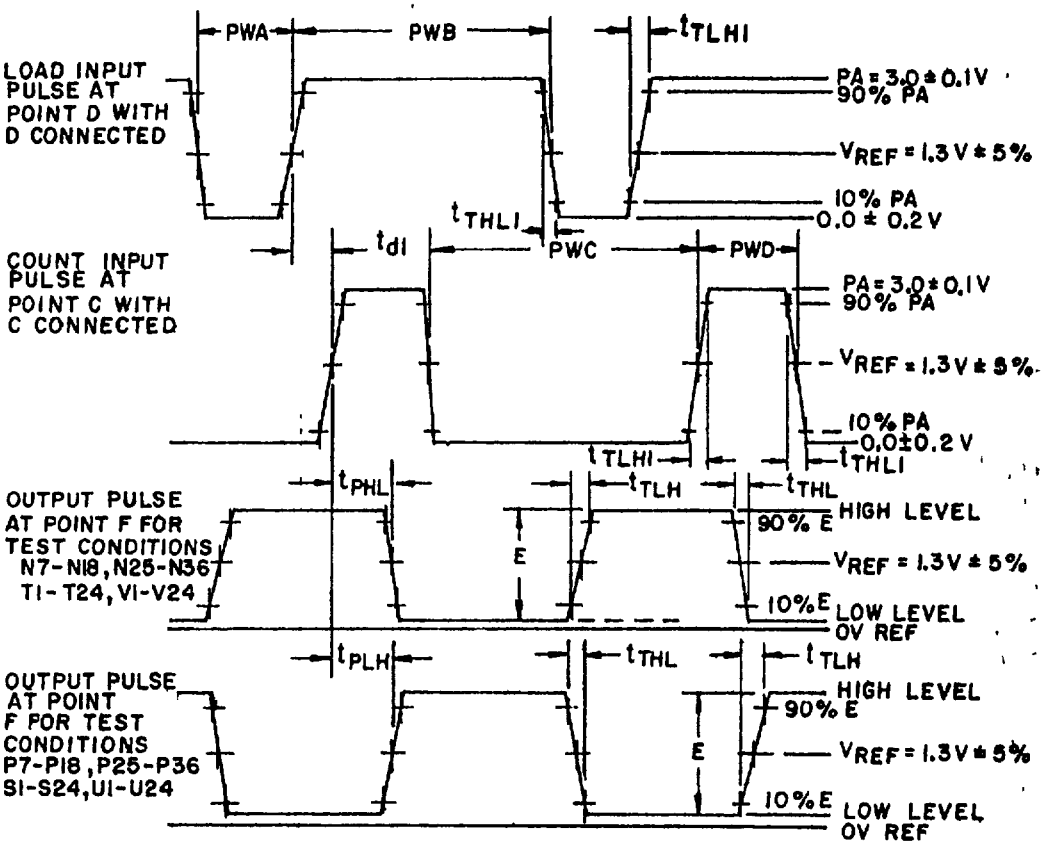


NOTES:

1. Input pulse parameters:
   PWA = 100 ±10 ns.
   PWB = 900 ±90 ns.
   PWC = 400 ±40 ns.
   PWD = 100 ±10 ns.
   t_{dl} = 100 ±10 ns.
   $t_{TLHI} = t_{THLI}$ = 10 ±2 ns.
2. Verify that the output low level is 0.4 V maximum and the output high level is 2.5 V minimum
3. Waveform at point C is obtained by placing item J01-1 in the double pulse mode.
4. All voltage levels are referenced to ground.

*FIGURE 16. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PHL} and t_{PLH}) test circuit waveforms for tables XIII and XIV.

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NOTES:
1. Input pulse parameters:
   PWA = 100 ±10 ns.
   PWB = 900 ±90 ns.
   PWC = 900 ±90 ns.
P PWD = 100 ±10 ns.
   t_{d1} = 100 ±10 ns.
   $t_{TLH1} = t_{THL1}$ = 10 ±2 ns.
2. Verify that all output low levels are 0.4 V maximum and all output high levels are 2.5 V minimum.
3. All voltage levels are referenced to ground.

* FIGURE 17. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XIII and XV.

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*TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test and minimum clear pulse width check.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
W1	Propagation delay time (t_{PLH}), load to QA1, C_L = 50 pF	15,16, 23,40	Open	14	21	38	4	44	ns	18,19
X1	Propagation delay time (t_{PHL}), load to QA1, C_L = 50 pF	15,16, 23,40		14		38				
W2	Propagation delay time (t_{PLH}), load to QA2, C_L = 50 pF	8,9, 23,35		13		33				
X2	Propagation delay time (t_{PHL}), load to QA2, C_L = 50 pF	8,9, 23,35		13		33				
W3	Propagation delay time (t_{PLH}), load to QA3, C_L = 50 pF	2,3, 23,24		5		27				
X3	Propagation delay time (t_{PHL}), load to QA3, C_L = 50 pF	2,3, 23,24		5		27				
W4	Propagation delay time (t_{PLH}), load to QB1, C_L = 50 pF	14,15, 16,23		40		37				
X4	Propagation delay time (t_{PHL}), load to QB1, C_L = 50 pF	14,15, 16,23		40		37				
W5	Propagation delay time (t_{PLH}), load to QB2, C_L = 50 pF	8,9, 13,23		35		34				
X5	Propagation delay time (t_{PHL}), load to QB2, C_L = 50 pF	8,9, 13,23		35		34				

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*TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test and minimum clear pulse width check - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
W6	Propagation delay time (t_{PLH}), load to QB3, C_L = 50 pF	2,3, 5,23	Open	24	21	28	4	44	ns	18,19
X6	Propagation delay time (t_{PHL}), load to QB3, C_L = 50 pF	2,3, 5,23		24		28				
W7	Propagation delay time (t_{PLH}), load to QC1, C_L = 50 pF	14,16, 23,40		15		18				
X7	Propagation delay time (t_{PHL}), load to QC1, C_L = 50 pF	14,16, 23,40		15		18				
W8	Propagation delay time (t_{PLH}), load to QC2, C_L = 50 pF	8,13, 23,35		9		30				
X8	Propagation delay time (t_{PHL}), load to QC2, C_L = 50 pF	8,13, 23,35		9		30				
W9	Propagation delay time (t_{PLH}), load to QC3, C_L = 50 pF	2,5, 23,24		3		6				
X9	Propagation delay time (t_{PHL}), load to QC3, C_L = 50 pF	2,5, 23,24		3		6				
W10	Propagation delay time (t_{PLH}), load to QD1, C_L = 50 pF	14,15, 23,40		16		19				
X10	Propagation delay time (t_{PHL}), load to QD1, C_L = 50 pF	14,15, 23,40		16		19				

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*TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test and minimum clear pulse width check - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
W11	Propagation delay time (t_{PLH}), load to QD2, C_L = 50 pF	9,13 23,35	Open	8	21	29	4	44	ns	18,19
X11	Propagation delay time (t_{PHL}), load to QD2, C_L = 50 pF	9,13, 23,35		8		29				
W12	Propagation delay time (t_{PLH}), load to QD3, C_L = 50 pF	3,5, 23,24		2		25				
X12	Propagation delay time (t_{PHL}), load to QD3, C_L = 50 pF	3,5, 23,24		2		25				
W13	Propagation delay time (t_{PLH}), load to QA1, C_L = 150 pF	15,16, 23,40		14		38	5	54		
X13	Propagation delay time (t_{PHL}), load to QA1, C_L = 150 pF	15,16, 23,40		14		38				
W14	Propagation delay time (t_{PLH}), load to QA2, C_L = 150 pF	8,9, 23,35		13		33				
X14	Propagation delay time (t_{PHL}), load to QA2, C_L = 150 pF	8,9, 23,35		13		33				
W15	Propagation delay time (t_{PLH}), load to QA3, C_L = 150 pF	2,3, 23,24		5		27				
X15	Propagation delay time (t_{PHL}), load to QA3, C_L = 150 pF	2,3, 23,24		5		27				

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*TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test and minimum clear pulse width check - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
W16	Propagation delay time (t_{PLH}), load to QB1, C_L = 150 pF	14,15, 16,23	Open	40	21	37	5	54	ns	18,19
X16	Propagation delay time (t_{PHL}), load to QB1, C_L = 150 pF	14,15, 16,23		40		37				
W17	Propagation delay time (t_{PLH}), load to QB2, C_L = 150 pF	8,9, 13,23		35		34				
X17	Propagation delay time (t_{PHL}), load to QB2, C_L = 150 pF	8,9, 13,23		35		34				
W18	Propagation delay time (t_{PLH}), load to QB3, C_L = 150 pF	2,3, 5,23		24		28				
X18	Propagation delay time (t_{PHL}), load to QB3, C_L = 150 pF	2,3, 5,23		24		28				
W19	Propagation delay time (t_{PLH}), load to QC1, C_L = 150 pF	14,16, 23,40		15		18				
X19	Propagation delay time (t_{PHL}), load to QC1, C_L = 150 pF	14,16, 23,40		15		18				
W20	Propagation delay time (t_{PLH}), load to QC2, C_L = 150 pF	8,13, 23,35		9		30				
X20	Propagation delay time (t_{PHL}), load to QC2, C_L = 150 pF	8,13, 23,35		9		30				

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*TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test and minimum clear pulse width check - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
W21	Propagation delay time (t_{PLH}), load to QC3, C_L = 150 pF	2,5, 23,24	Open	3	21	6	5	54	ns	18,19
X21	Propagation delay time (t_{PHL}), load to QC3, C_L = 150 pF	2,5, 23,24		3		6				
W22	Propagation delay time (t_{PLH}), load to QD1, C_L = 150 pF	14,15, 23,40		16		19				
X22	Propagation delay time (t_{PHL}), load to QD1, C_L = 150 pF	14,15, 23,40		16		19				
W23	Propagation delay time (t_{PLH}), load to QD2, C_L = 150 pF	9,13, 23,35		8		29				
X23	Propagation delay time (t_{PHL}), load to QD2, C_L = 150 pF	9,13, 23,35		8		29				
W24	Propagation delay time (t_{PLH}), load to QD3, C_L = 150 pF	3,5, 23,24		2		25				
X24	Propagation delay time (t_{PHL}), load to QD3, C_L = 150 pF	3,5, 23,24		2		25				
Y1	Propagation delay time (t_{PHL}), clear to QA1, C_L = 50 pF	15,16, 40	14	21	23	38	3	39		18,20
Y2	Propagation delay time (t_{PHL}), clear to QA2, C_L = 50 pF	8,9, 35	13			33				
Y3	Propagation delay time (t_{PHL}), clear to QA3, C_L = 50 pF	2,3, 24	5			27				
Y4	Propagation delay time (t_{PHL}), clear to QB1, C_L = 50 pF	14,15, 16	40			37				
Y5	Propagation delay time (t_{PHL}), clear to QB2, C_L = 50 pF	8,9, 13	35			34				

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*TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test and minimum clear pulse width check - Continued.

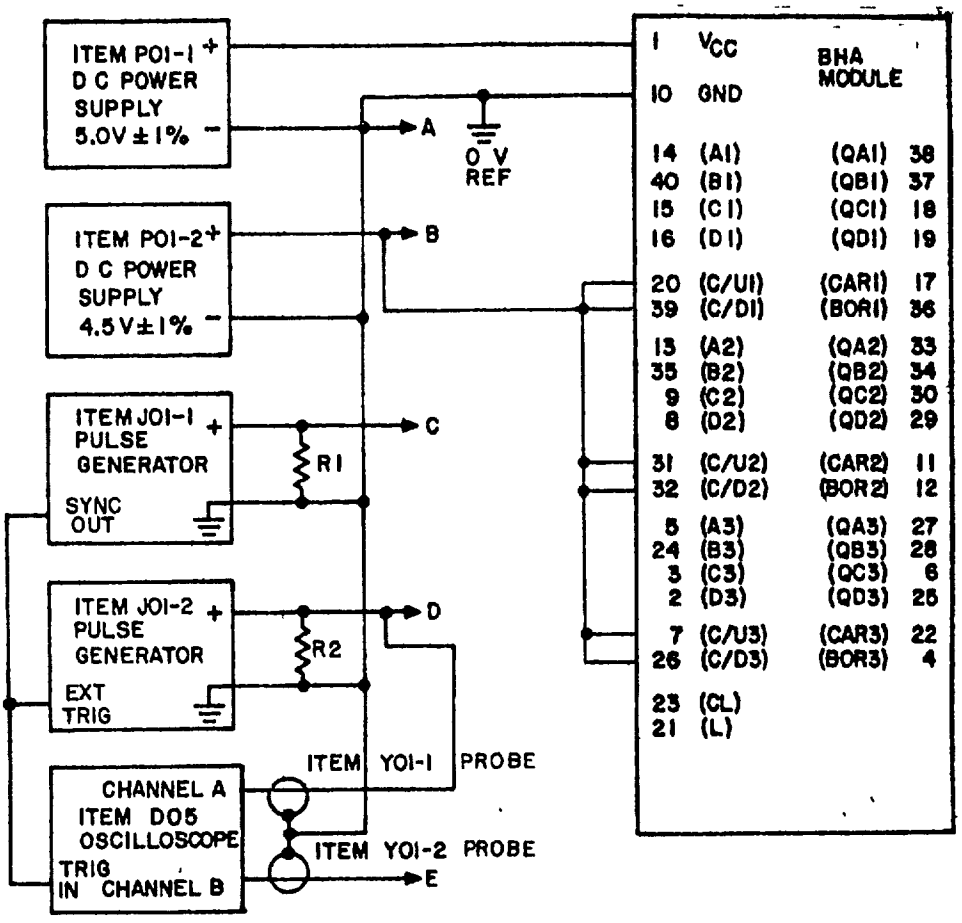
Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Y6	Propagation delay time (t_{PHL}), clear to QB3, C_L = 50 pF	2,3, 5,	24	21	23	28	3	39	ns	18,20
Y7	Propagation delay time (t_{PHL}), clear to QC1, C_L = 50 pF	14,16, 40	15			18				
Y8	Propagation delay time (t_{PHL}), clear to QC2, C_L = 50 pF	8,13, 35	9			30				
Y9	Propagation delay time (t_{PHL}), clear to QC3, C_L = 50 pF	2,5, 24	3			6				
Y10	Propagation delay time (t_{PHL}), clear to QD1, C_L = 50 pF	14,15, 40	16			19				
Y11	Propagation delay time (t_{PHL}), clear to QD2, C_L = 50 pF	9,13, 35	8			29				
Y12	Propagation delay time (t_{PHL}), clear to QD3, C_L = 50 pF	3,5, 24	2			25				
Y13	Propagation delay time (t_{PHL}), clear to QA1, C_L = 150 pF	15,16, 40	14			38	4	49		
Y14	Propagation delay time (t_{PHL}), clear to QA2, C_L = 150 pF	8,9, 35	13			33				
Y15	Propagation delay time (t_{PHL}), clear to QA3, C_L = 150 pF	2,3, 24	5			27				

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*TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test and minimum clear pulse width check - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Y16	Propagation delay time (t_{PHL}), clear to QB1, C_L = 150 pF	14,15, 16	40	21	23	37	4	49	ns	18,20
Y17	Propagation delay time (t_{PHL}), clear to QB2, C_L = 150 pF	8,9, 13	35			34				
Y18	Propagation delay time (t_{PHL}), clear to QB3, C_L = 150 pF	2,3, 5	24			28				
Y19	Propagation delay time (t_{PHL}), clear to QC1, C_L = 150 pF	14,16, 40	15			18				
Y20	Propagation delay time (t_{PHL}), clear to QC2, C_L = 150 pF	8,13, 35	9			30				
Y21	Propagation delay time (t_{PHL}), clear to QC3, C_L = 150 pF	2,5, 24	3			6				
Y22	Propagation delay time (t_{PHL}), clear to QD1, C_L = 150 pF	14,15, 40	16			19				
Y23	Propagation delay time (t_{PHL}), clear to QD2, C_L = 150 pF	9,13, 35	8			29				
Y24	Propagation delay time (t_{PHL}), clear to QD3, C_L = 150 pF	3,5, 24	2			25				

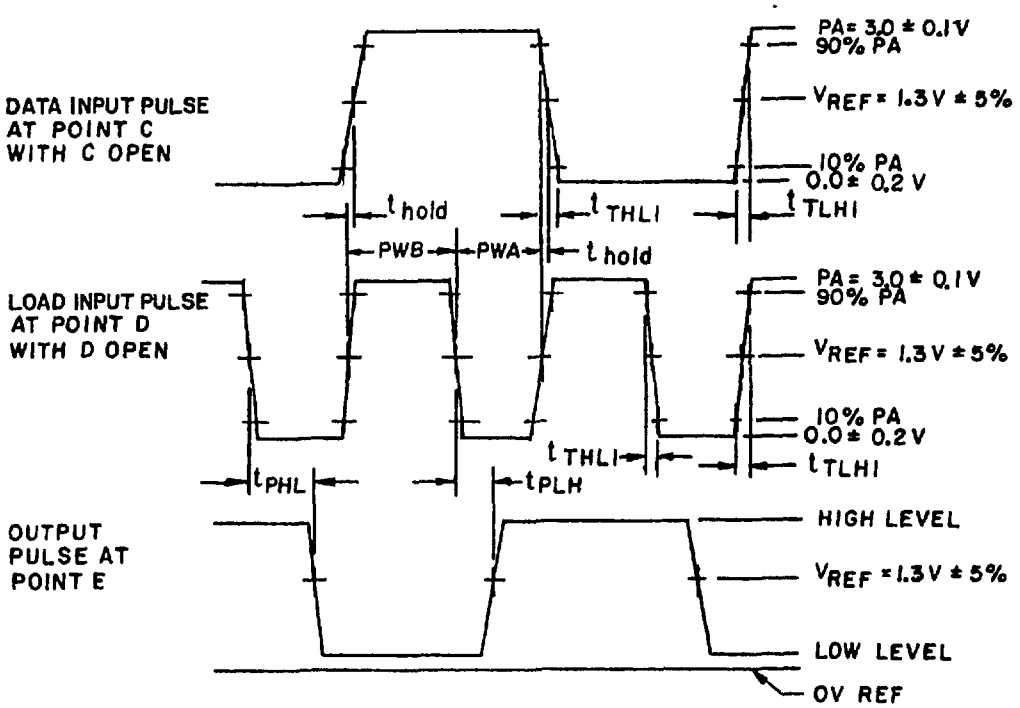
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. Load all outputs with load as specified on figure 12.

* FIGURE 18. Propagation delay time (t_{PLH} and t_{PHL}) test and minimum clear pulse width check test circuit for table XVI.

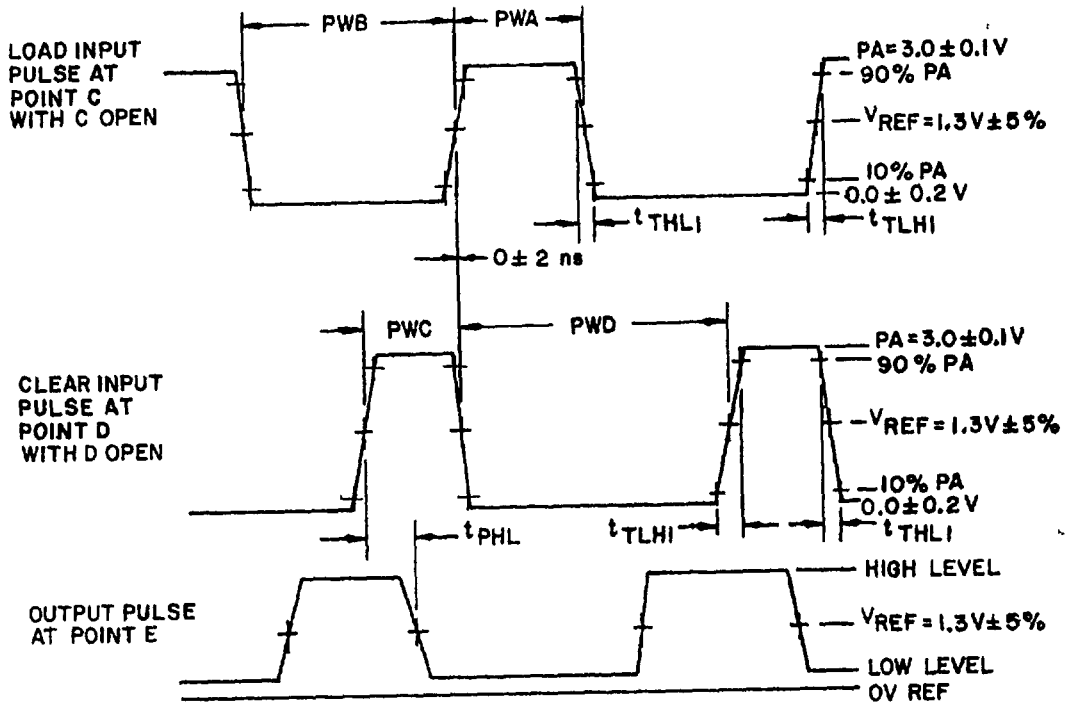
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NOTES:
1. Input pulse parameters:
   PWA = 100 ±10 ns.
   PWB = 100 ±10 ns.
   t_{hold} = 10 ±2 ns.
   $t_{TLH1} = t_{THL1}$ = 6 ±2 ns.
2. Verify that the output low level is 0.4 V maximum and the output high level is 2.5 V minimum.
3. All voltage levels are referenced to ground.

* FIGURE 19. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XVI.

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NOTES:
1. Input pulse parameters:
   PWA = 40 ±4 ns.
   PWB = 60 ±6 ns.
   PWC = 20 $^{+2}_{-0}$ ns (clear pulse width).
   PWD = 80 ±8 ns.
   $t_{TLHI} = t_{THLI}$ = 6 ±2 ns.
2. Verify that the output low level is 0.4 V maximum and the output high level is 2.5 V minimum.
3. All voltages levels are referenced to ground.

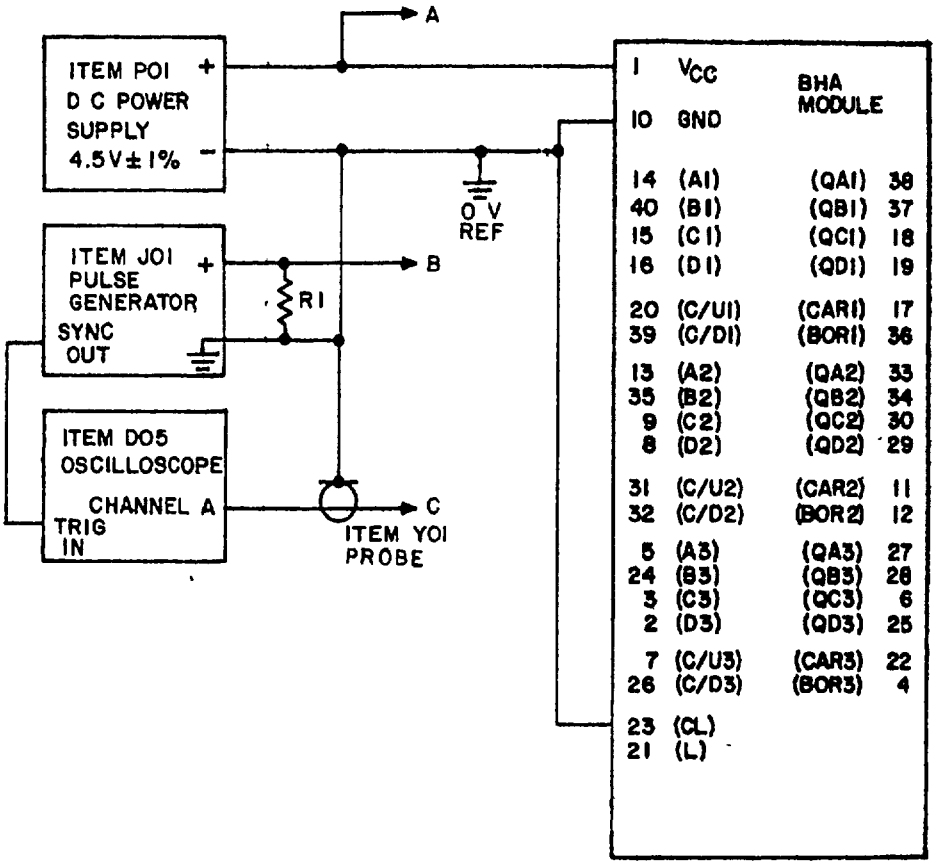
* FIGURE 20. Propagation delay time (t_{PLH} and t_{PHL}) and minimum clear pulse width check test circuit waveforms for table XVI.

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*TABLE XVII. Maximum counting frequency check (minimum count pulse width check) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Z1	Maximum counting frequency check (minimum count pulse width check), carry period	39, 21	20	17	844	948	ns	21, 22
Z2	Maximum counting frequency check (minimum count pulse width check), carry period	32, 21	31	11				
Z3	Maximum counting frequency check (minimum count pulse width check), carry period	26, 21	7	22				
Z4	Maximum counting frequency check (minimum count pulse width check), borrow period	20, 21	39	36				
Z5	Maximum counting frequency check (minimum count pulse width check), borrow period	31, 21	32	12				
Z6	Maximum counting frequency check (minimum count pulse width check), borrow period	7, 21	26	4				

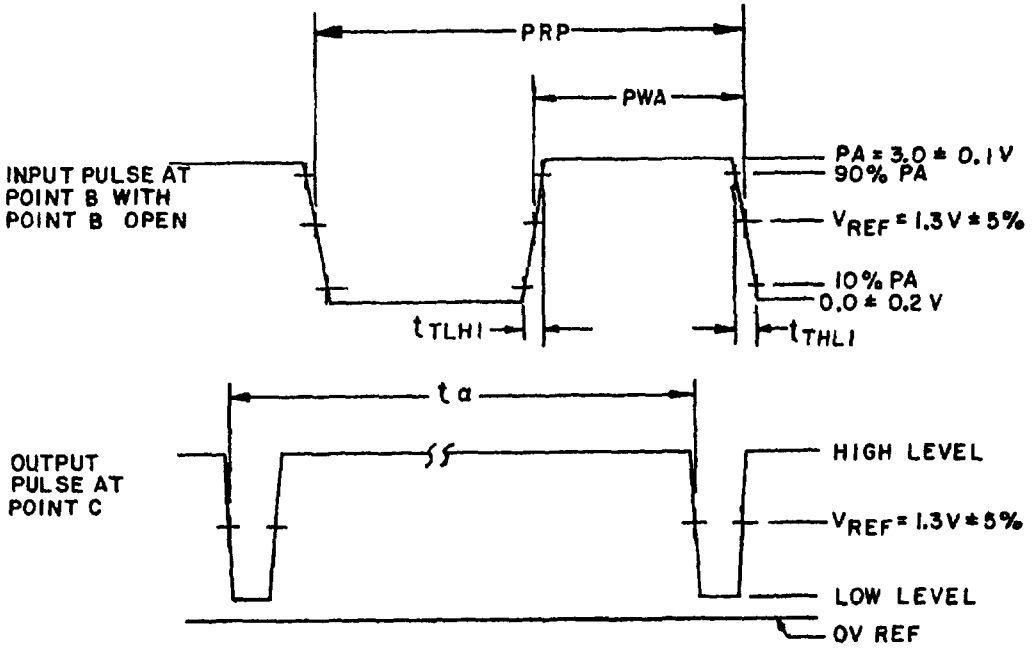
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NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. Load all outputs with the load as specified on figure 12.

FIGURE 21. Maximum counting frequency check (minimum count pulse width check) test circuit for table XVII.

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NOTES:
1. Input pulse parameters:
   PWA = 27 +2, -0 ns.
   PRP = 54 +4, -0 ns.
   $t_{TLH1} = t_{THL1} = 6 \pm 2$ ns.
2. Verify that the output low level is 0.4 V maximum and the output high level is 2.5 V minimum.
3. All voltage levels are referenced to ground.
4. t_a = CARRY or BORROW period.

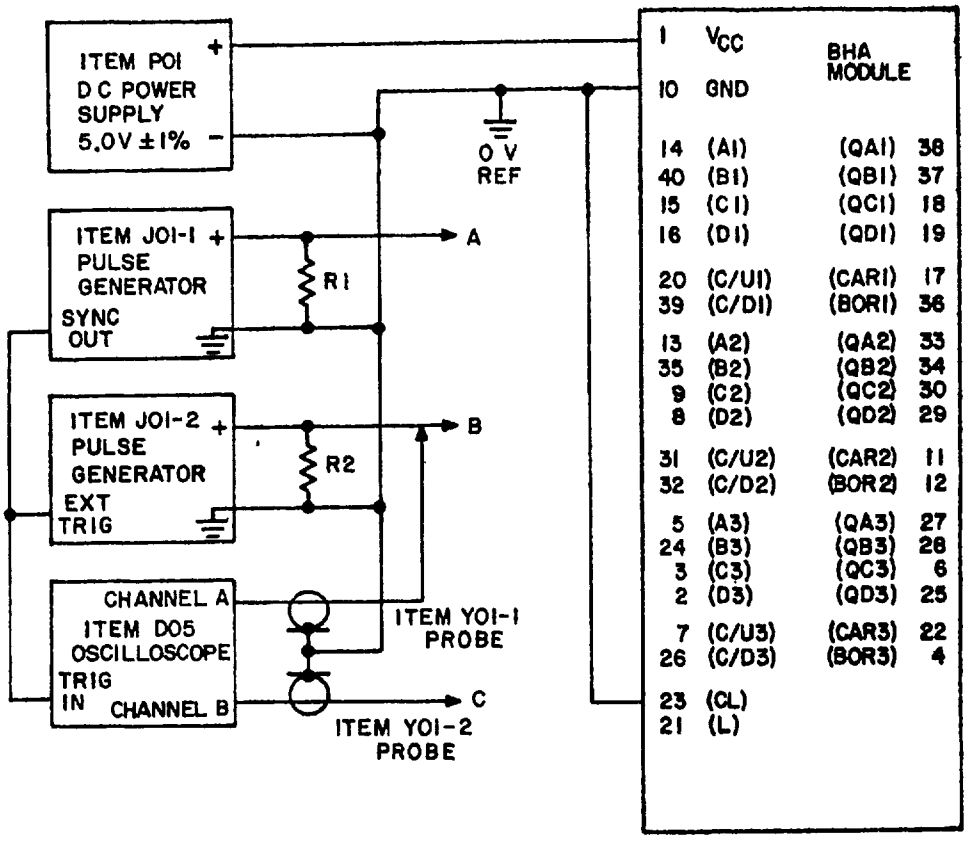
* FIGURE 22. Maximum counting frequency check (minimum count pulse width check) test circuit waveforms for table XVII.

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TABLE XVIII. Setup time (data to load) and minimum load pulse width check test.

Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
AA1	QA1	Setup time (data to load) and minimum load pulse width check	14	21	38	110	130	ns	23,24
AA2	QA2	↓	13	↓	33	↓	↓	↓	↓
AA3	QA3	↓	5	↓	27	↓	↓	↓	↓
AA4	QB1	↓	40	↓	37	↓	↓	↓	↓
AA5	QB2	↓	35	↓	34	↓	↓	↓	↓
AA6	QB3	↓	24	↓	28	↓	↓	↓	↓
AA7	QC1	↓	15	↓	18	↓	↓	↓	↓
AA8	QC2	↓	9	↓	30	↓	↓	↓	↓
AA9	QC3	↓	3	↓	6	↓	↓	↓	↓
AA10	QD1	↓	16	↓	19	↓	↓	↓	↓
AA11	QD2	↓	8	↓	29	↓	↓	↓	↓
AA12	QD3	↓	2	↓	25	↓	↓	↓	↓

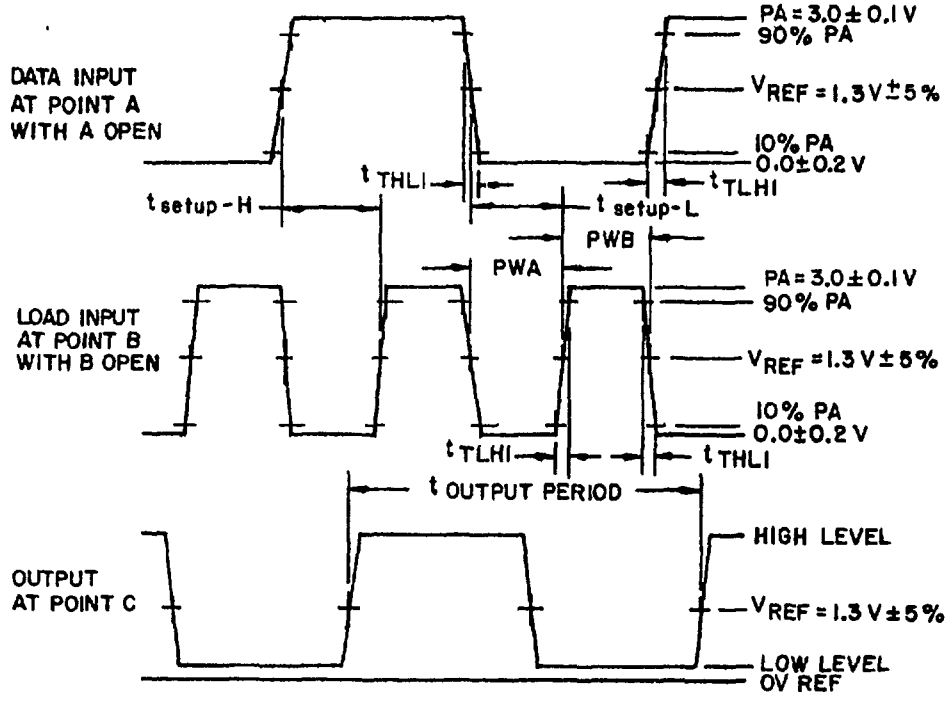
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. Load all outputs with the load as specified on figure 12.

*FIGURE 23. Setup time (data to load) and minimum load pulse width check test circuit for table XVIII.

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NOTES:
1. Input pulse parameters:

   PWA = $20\ ^{+2}_{-0}$ ns.
   PWB = 40 ±5 ns.
   $t_{setup-H} = t_{setup-L} = 20\ ^{+2}_{-0}$ ns.
   $t_{TLHH} = t_{THLL} = 6 \pm 2$ ns.
2. Verify that the output low level is 0.4 V maximum and the output high level is 2.5 V minimum.
3. All voltages levels are referenced to ground.

FIGURE 24. Setup time (data to load) and minimum load pulse width check test circuit waveforms for table XVIII.

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TABLE XIX. Output crosstalk (high-level and low-level and duration) test.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
BB1	QA1 Output crosstalk, high level	14	Open	21	2,3,5,8, 9,13,15, 16,24, 35,40	38	0	400	mV	25,26
BB2	QB1	40			2,3,5,8, 9,13,14, 15,16, 24,35	37				
BB3	QC1	15			2,3,5,8, 9,13,14, 16,24, 35,40	18				
BB4	QD1	16			2,3,5,8, 9,13,14, 15,24, 35,40	19				
BB5	QA2	13			2,3,5,8, 9,14,15, 16,24, 35,40	33				
BB6	QB2	35			2,3,5,8, 9,13,14, 15,16, 24,40	34				
BB7	QC2	9			2,3,5,8, 13,14, 15,16, 24,35,40	30				
BB8	QD2	8			2,3,5,9, 13,14, 15,16, 24,35,40	29				

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TABLE XIX. Output crosstalk (high-level and low-level and duration) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
BB9	QA3 Output crosstalk, high level	5	Open	21	2,3,8,9, 13,14, 15,16, 24,35,40	27	0	400	mV	25, 26
BB10	QB3	24			2,3,5,8, 9,13,14, 15,16, 35,40	28				
BB11	QC3	3			2,5,8,9, 13,14, 15,16, 24,35,40	6				
BB12	QD3	2			3,5,8,9, 13,14, 15,16, 24,35,40	25				
CC1	QA1 Output crosstalk, low level	Open	14		2,3,5,8, 9,13,15, 16,24, 35,40	38		500		
CC2	QB1		40		2,3,5,8, 9,13,14, 15,16, 24,35	37				
CC3	QC1		15		2,3,5,8, 9,13,14, 16,24, 35,40	18				
CC4	QD1		16		2,3,5,8, 9,13,14, 15,24, 35,40	19				

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TABLE XIX. Output crosstalk (high-level and low-level and duration) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
CC5	QA2 Output crosstalk, low level	Open	13	21	2,3,5,8, 9,14,15, 16,24, 35,40	33	0	500	mV	25, 26
CC6	QB2		35		2,3,5,8, 9,13,14, 15,16, 24,40	34				
CC7	QC2		9		2,3,5,8, 13,14, 15,16, 24,35,40	30				
CC8	QD2		8		2,3,5,9, 13,14, 15,16, 24,35,40	29				
CC9	QA3		5		2,3,8,9, 13,14, 15,16, 24,35,40	27				
CC10	QB3		24		2,3,5,8, 9,13,14, 15,16, 35,40	28				
CC11	QC3		3		2,5,8,9, 13,14, 15,16, 24,35,40	6				
CC12	QD3		2		3,5,8,9, 13,14, 15,16, 24,35,40	25				

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TABLE XIX. Output crosstalk (high-level and low-level and duration) test - Continued.

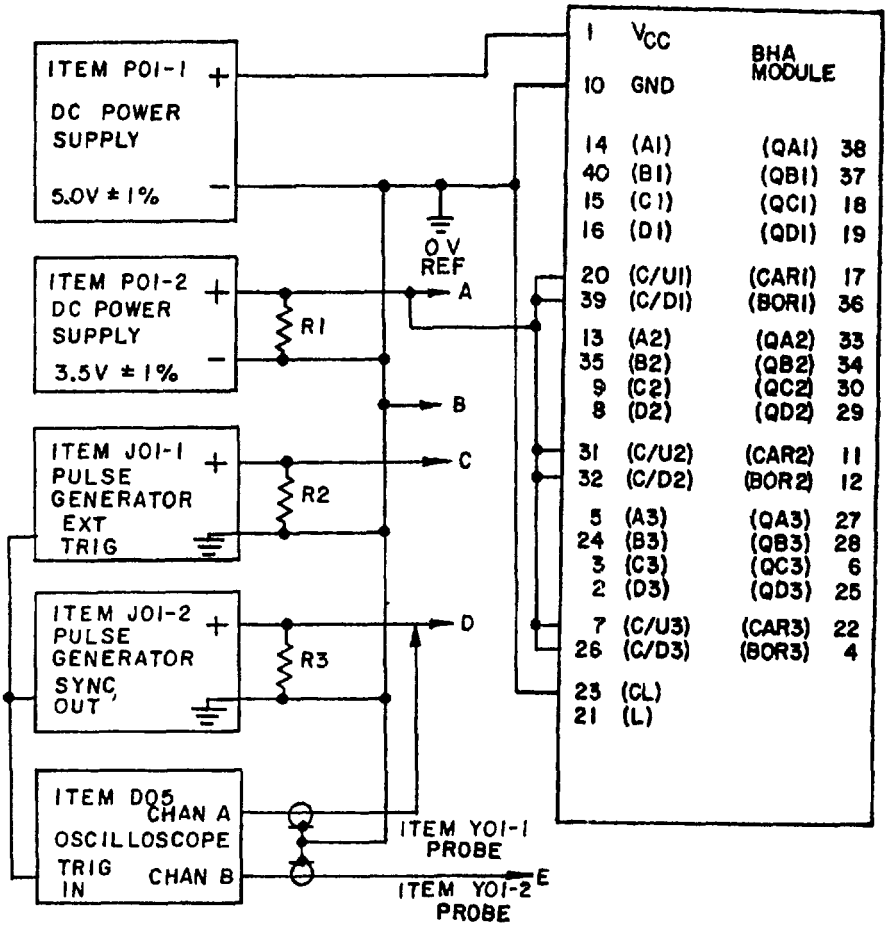
Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
CC13	QA1 Output crosstalk, duration	Open	14	21	2,3,5,8, 9,13,15, 16,24, 35,40	38	0	400	mV	25, 27
CC14	QB1		40		2,3,5,8, 9,13,14, 15,16, 24,35	37				
CC15	QC1		15		2,3,5,8, 9,13,14, 16,24, 35,40	18				
CC16	QD1		16		2,3,5,8, 9,13,14, 15,24, 35,40	19				
CC17	QA2		13		2,3,5,8, 9,14,15, 16,24, 35,40	33				
CC18	QB2		35		2,3,5,8, 9,13,14, 15,16, 24,40	34				
CC19	QC2		9		2,3,5,8, 13,14, 15,16, 24,35,40	30				
CC20	QD2		8		2,3,5,9, 13,14, 15,16, 24,35,40	29				

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TABLE XIX. Output crosstalk (high-level and low-level and duration) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
CC21	QA3 Output crosstalk, duration	Open	5	21	2,3,8,9, 13,14, 15,16, 24,35,40	27	0	400	mV	25, 27
CC22	QB3		24		2,3,5,8, 9,13,14, 15,16, 35,40	28				
CC23	QC3		3		2,5,8,9, 13,14, 15,16, 24,35,40	6				
CC24	QD3		2		3,5,8,9, 13,14, 15,16, 24,35,40	25				

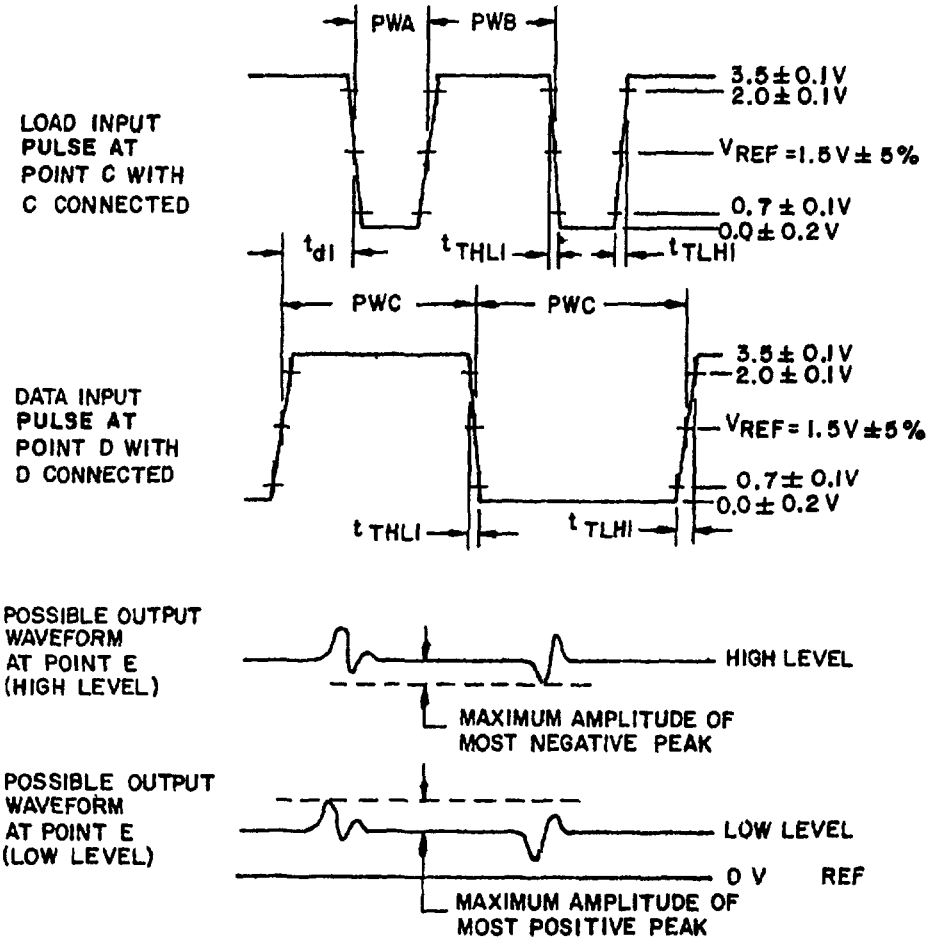
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NOTES:
1. R1-R3 = 51Ω ±5%, 1/2 W.
2. Load all outputs with the load as specified on figure 12.

FIGURE 25. Output crosstalk (high-level and low-level, and duration) test circuit for table XIX.

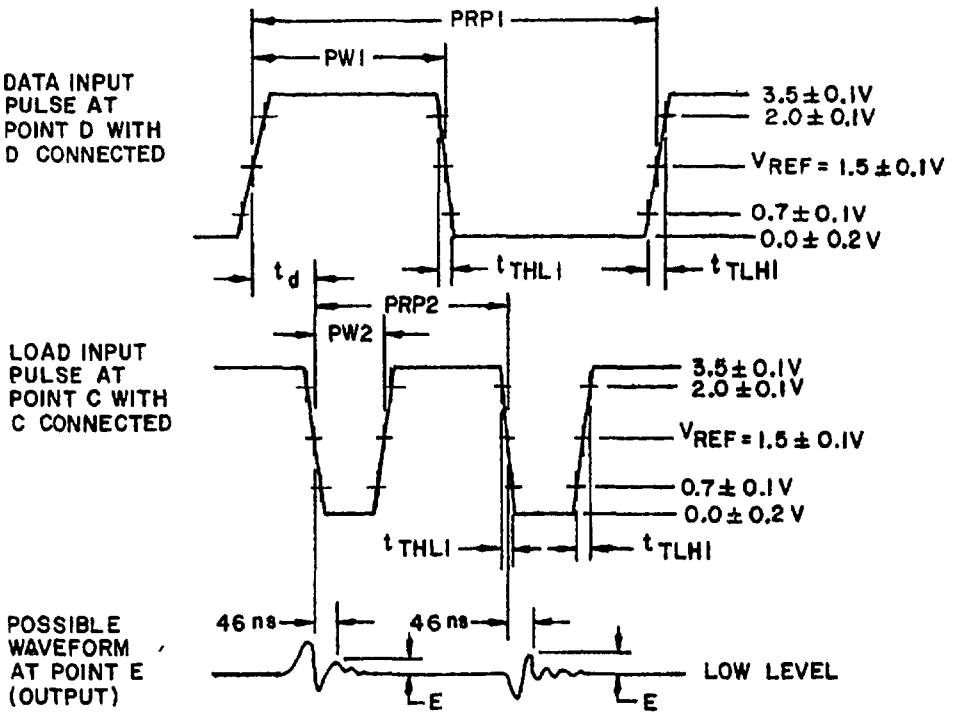
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:25Z
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NOTE: Input pulse parameters:
PWA = 75 ±5 ns.
PWB = 175 ±10 ns.
PWC = 250 ±25 ns.
$t_{TLH1} = t_{THL1} = 3.5 \pm 1.5$ ns (verified at each input pin).
$t_{d1} = 85 \pm 5$ ns.

*FIGURE 26. Output crosstalk (high-level and low-level) test circuit waveforms table XIX.

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NOTES:
1. Input pulse parameters;
   PRP1 = 500 ±50 ns.
   PW1 = 250 ±25 ns.
   PRP2 = 250 ±25 ns.
   PW2 = 75 ±5 ns.
   t_d = 85 ±5 ns.
   $t_{TLH1} = t_{THL1}$ = 3.5 ±1.5 ns (verified at each input pin).
2. E is the maximum amplitude of the most positive peak measured after a settle time of 46 ns has elapsed after any negative transition of the load pulse.

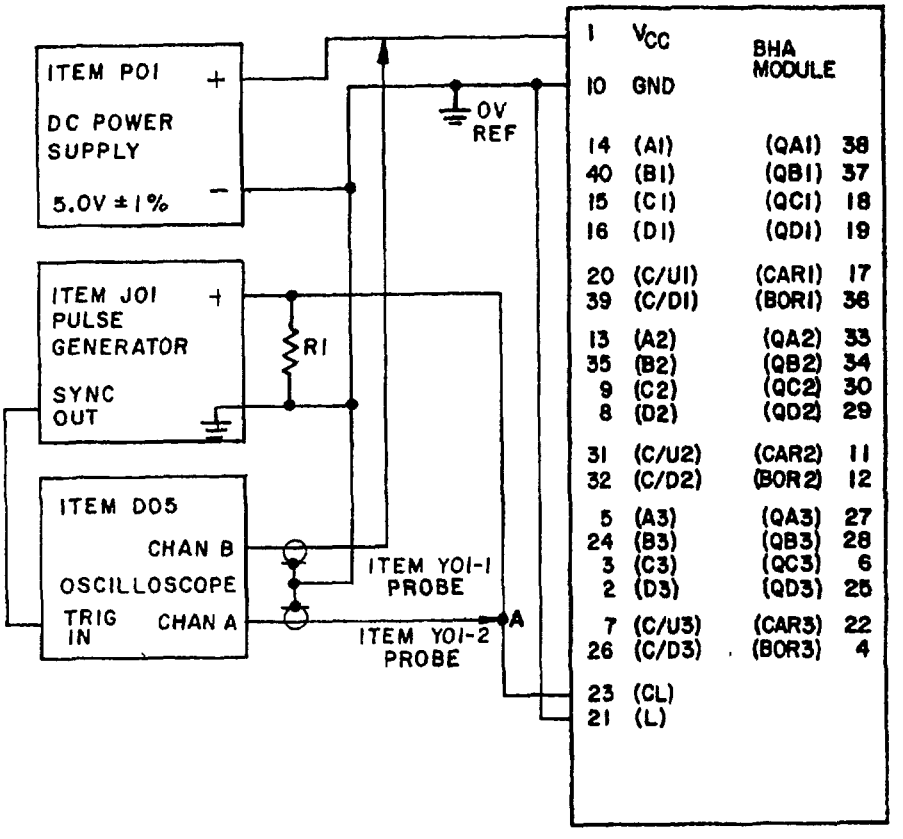
*FIGURE 27. Output crosstalk (duration) test circuit waveforms for table XIX.

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TABLE XX. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
DD	Power supply transient amplitude	0	400	mV	28, 29

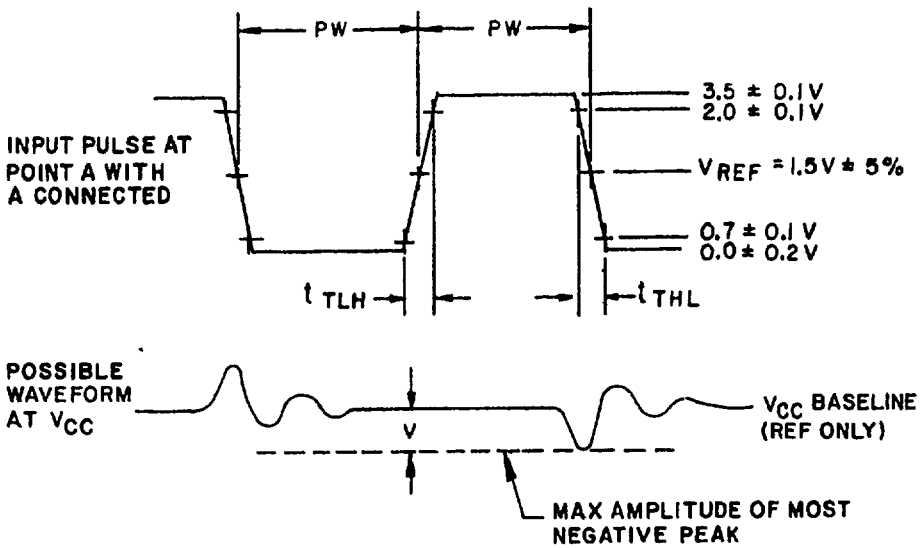
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NOTE: R1 = 51Ω ±5%, 1/2 W.

FIGURE 28. Power supply transient amplitude test circuit for table XX.

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NOTES:
1. Input pulse parameters:
   PW = 500 ±50 ns.
   $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns.
2. V = Measurement is made from the base line to the most negative excursion of the waveform.

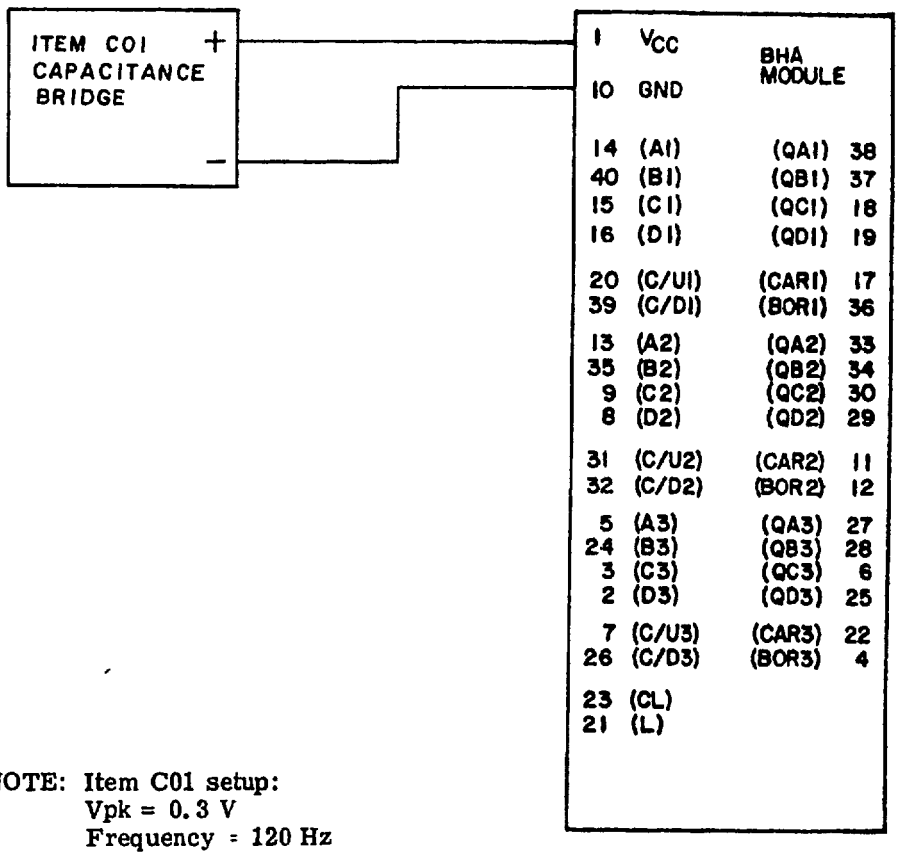
* FIGURE 29. Power supply transient amplitude test circuit waveforms for table XX.

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TABLE XXI. Power supply filter capacitance test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
EE	Power supply filter capacitance	1.6	3.0	µF	30

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NOTE: Item C01 setup:
Vpk = 0.3 V
Frequency = 120 Hz

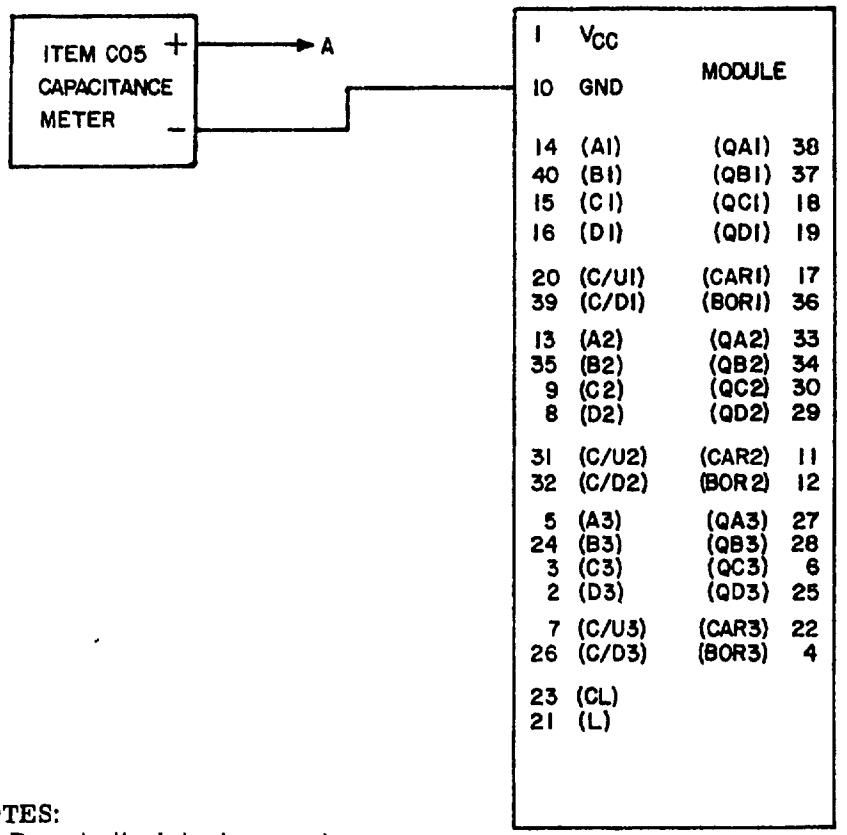
FIGURE 30. Power supply filter capacitance test circuit for table XXI.

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TABLE XXII. Input capacitance test.

Test condition	Parameter		Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
FF1	A1	Input capacitance	14	2	15	pF	31
FF2	B1		40				
FF3	C1		15				
FF4	D1		16				
FF5	C/U1		20				
FF6	C/D1		39				
FF7	A2		13				
FF8	B2		35				
FF9	C2		9				
FF10	D2		8				
FF11	C/U2		31				
FF12	C/D2		32				
FF13	A3		5				
FF14	B3		24				
FF15	C3		3				
FF16	D3		2				
FF17	C/U3		7				
FF18	C/D3		26				
FF19	CL		23	4	30		
FF20	L		21	4	30		

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NOTES:
1. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
2. Null out the effects of the test leads prior to connecting the module.

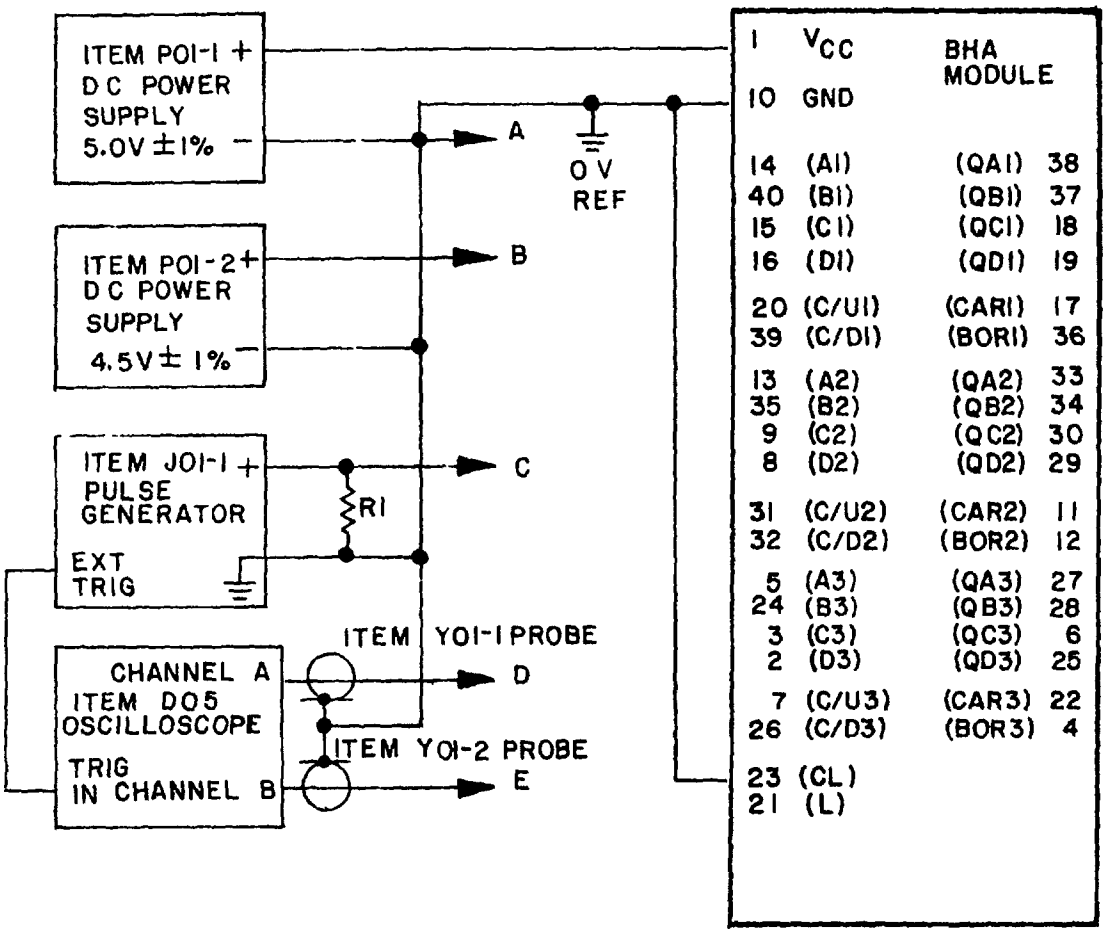
FIGURE 31. Input capacitance test cirucit for table XXII.

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TABLE XXIII. Borrow pulse width test.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
GG1	Borrow pulse width	14,15, 16,40	7,20,21, 31	39	39	36	300	500	ns	32, 33
GG2	Borrow pulse width	8,9,13, 35	7,20,21, 31	32	32	12	300	500	ns	32, 33
GG3	Borrow pulse width	2,3,5, 24	7,20,21, 31	26	26	4	300	500	ns	32, 33

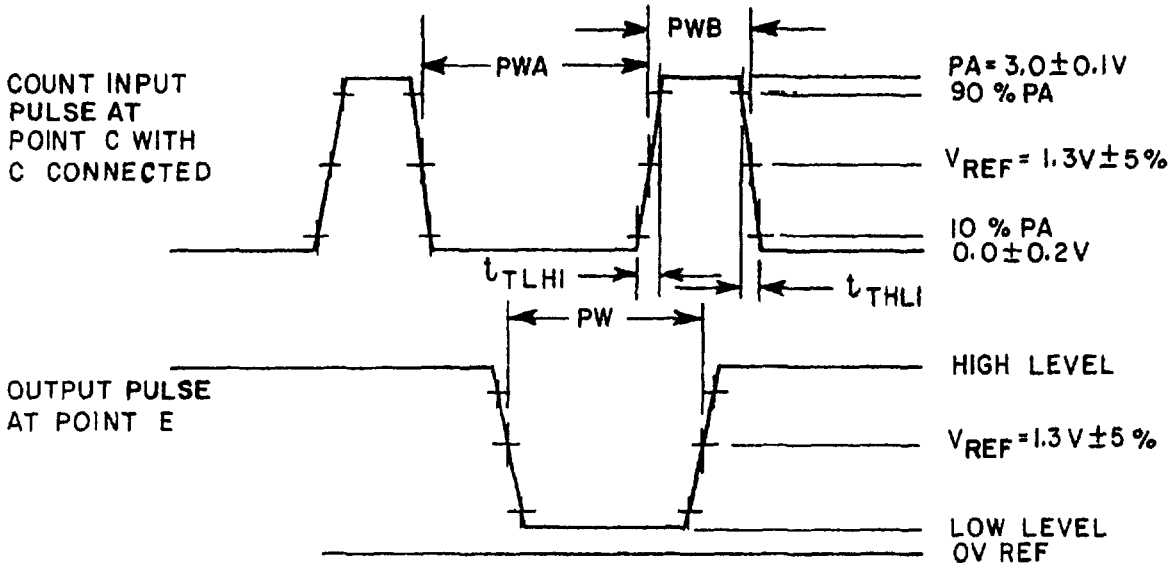
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NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. Load all outputs with load as specified on figure 12.

*FIGURE 32. Borrow pulse width test circuit for table XXIII.

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NOTES:
1. Input pulse parameters.
   PWA = 400 ±40 ns.
   PWB = 100 ±10 ns.
   $t_{TLH1} = t_{THL1} = 10 \pm 2$ ns.
2. Verify that the output low level is 0.4 V maximum and the output high level is 2.5 V minimum.
3. Pulse at point C is obtained by placing pulse generator J01-1 in the double pulse mode.
4. All voltage levels are referenced to ground.

* FIGURE 33. Borrow pulse width test circuit waveforms for table XXIII.

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TABLE XXIV. Minimum load hold time test.

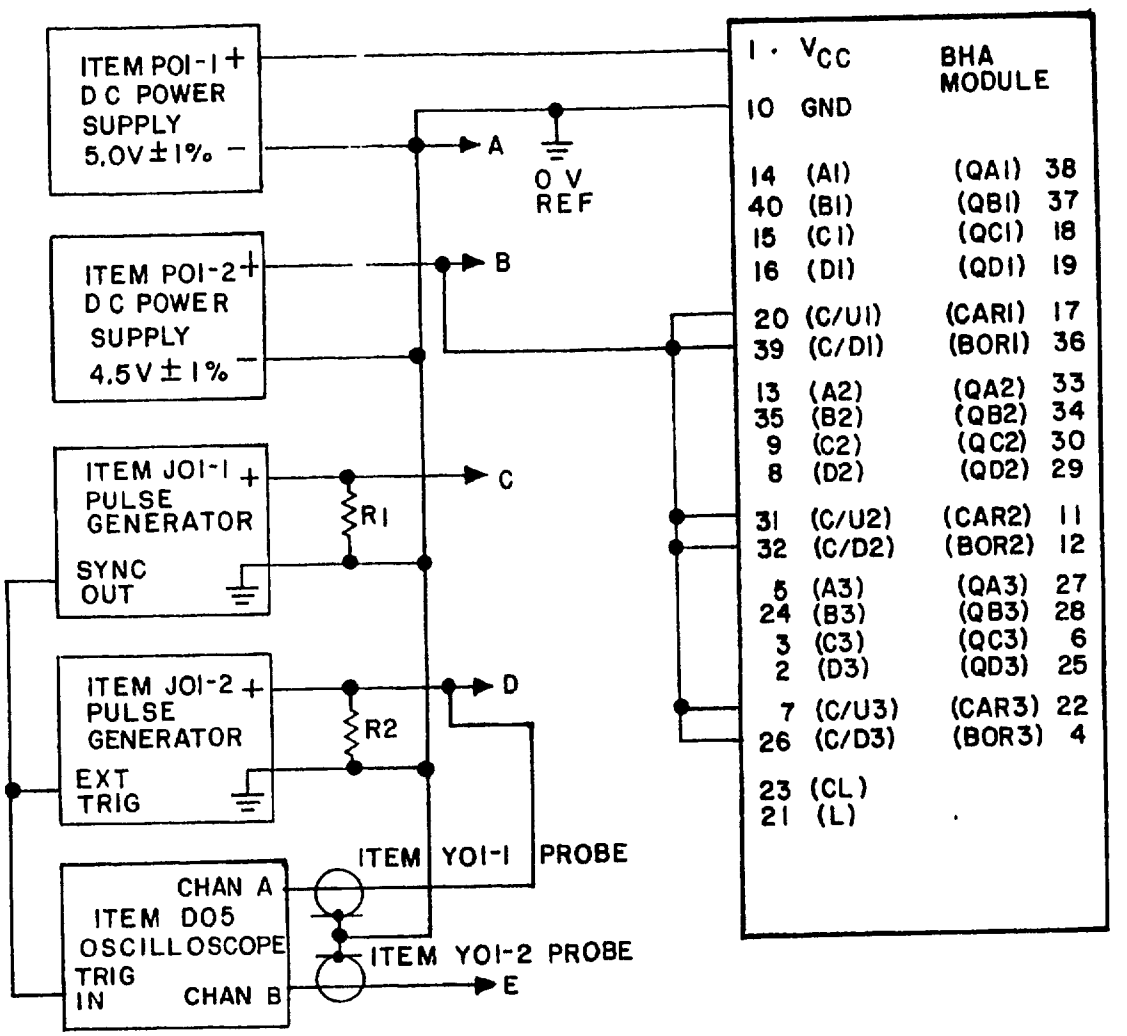
Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
HH1	Minimum load hold time (t_{H1}), C_L = 50 pF	15,16, 23,40	Open	14	*21	38	170	230	ns	34, 35
HH2	Minimum load hold time (t_{H0}), C_L = 50 pF	15,16, 23,40		14		38				
HH3	Minimum load hold time (t_{H1}), C_L = 50 pF	8,9, 23,35		13		33				
HH4	Minimum load hold time (t_{H0}), C_L = 50 pF	8,9, 23,35		13		33				
HH5	Minimum load hold time (t_{H1}), C_L = 50 pF	2,3, 23,24		5		27				
HH6	Minimum load hold time (t_{H0}), C_L = 50 pF	2,3, 23,24		5		27				
HH7	Minimum load hold time (t_{H1}), C_L = 50 pF	14,15, 16,23		40		37				
HH8	Minimum load hold time (t_{H0}), C_L = 50 pF	14,15, 16,23		40		37				
HH9	Minimum load hold time (t_{H1}), C_L = 50 pF	8,9, 13,23		35		34				
HH10	Minimum load hold time (t_{H0}), C_L = 50 pF	8,9, 13,23		35		34				
HH11	Minimum load hold time (t_{H1}), C_L = 50 pF	2,3, 5,23		24		28				
HH12	Minimum load hold time (t_{H0}), C_L = 50 pF	2,3, 5,23		24		28				
HH13	Minimum load hold time (t_{H1}), C_L = 50 pF	14,16, 23,40		15		18				
HH14	Minimum load hold time (t_{H0}), C_L = 50 pF	14,16, 23,40		15		18				

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TABLE XXIV. Minimum load hold time test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
HH15	Minimum load hold time (t_{H1}), C_L = 50 pF	8,13, 23,35	Open	9	21	30	170	230	ns	34, 35
HH16	Minimum load hold time (t_{H0}), C_L = 50 pF	8,13, 23,35	↓	9	↓	30	↓	↓	↓	↓
HH17	Minimum load hold time (t_{H1}), C_L = 50 pF	2,5, 23,24	↓	3	↓	6	↓	↓	↓	↓
HH18	Minimum load hold time (t_{H0}), C_L = 50 pF	2,5, 23,24	↓	3	↓	6	↓	↓	↓	↓
HH19	Minimum load hold time (t_{H1}), C_L = 50 pF	14,15, 23,40	↓	16	↓	19	↓	↓	↓	↓
HH20	Minimum load hold time (t_{H0}), C_L = 50 pF	14,15, 23,40	↓	16	↓	19	↓	↓	↓	↓
HH21	Minimum load hold time (t_{H1}), C_L = 50 pF	9,13, 23,35	↓	8	↓	29	↓	↓	↓	↓
HH22	Minimum load hold time (t_{H0}), C_L = 50 pF	9,13, 23,35	↓	8	↓	29	↓	↓	↓	↓
HH23	Minimum load hold time (t_{H1}), C_L = 50 pF	3,5, 23,24	↓	2	↓	25	↓	↓	↓	↓
HH24	Minimum load hold time (t_{H0}), C_L = 50 pF	3,5, 23,24	↓	2	↓	25	↓	↓	↓	↓

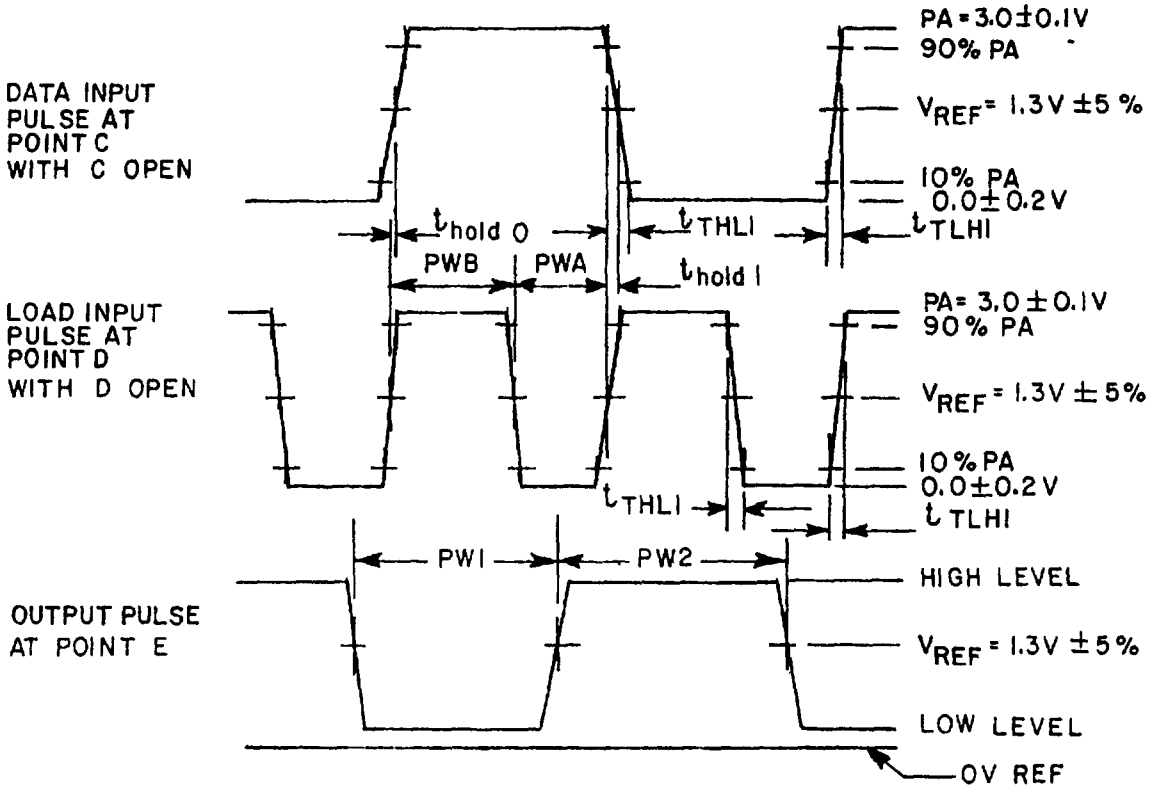
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NOTES.
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. Load all outputs with load as specified on figure 12.

* FIGURE 34. Minimum load hold time test circuit for table XXIV.

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NOTES
1. Input pulse parameters:
   PWA = 100 ±10 ns
   PWB = 100 ±10 ns.
   $t_{TLH1} = t_{THL1} = 6 \pm 2$ ns
2. Verify that the output low level is 0.4 V maximum and the output high level is 2.5 V minimum.
3. All voltage levels are referenced to ground
4. $t_{hold1} = 5\ +2, -0$ ns, $t_{hold0} = 10 \pm 2$ ns for test conditions HH1-HH23 (odd).
5. $t_{hold1} = 10 \pm 2$ ns, $t_{hold0} = 5\ +2, -0$ ns for test conditions HH2-HH24 (even).
6. Measure PW1 for test conditions HH2-HH24 (even).
7. Measure PW2 for test conditions HH1-HH23 (odd).

* FIGURE 35. Minimum load hold time test circuit waveforms for table XXIV.

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